Concert Management and Musical Instrument Trading Platform

SE-3105 Software Project Lab 2

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1.Introduction

1.1 Purpose of the Document

This Software Requirements Specification (SRS) document provides a comprehensive description of the functional and non-functional requirements for GaanBajna, a multi-role live concert and artist marketplace platform. This document serves as a formal agreement between stakeholders, developers, and the project team, ensuring a common understanding of system capabilities, constraints, and features. It will guide the design, development, testing, and implementation phases of the project.

1.2 Scope of the Project

GaanBajna is a web-based platform designed to streamline the live music event ecosystem by connecting four primary user roles: **Audience, Singers, Organizers,** and **Admin**. The system facilitates concert booking, ticket sales, merchandise marketplace, and artist management through a secure, role-based interface.

In-Scope Features:

- Multi-role registration with email verification and admin approval workflow
- Singer calendar management and fixed-fee concert booking system
- Integrated ticketing with QR-based entry validation
- Marketplace for artist and organizer merchandise
- Comprehensive admin panel for user management and moderation
- Payment gateway integration for financial transactions
- Real-time ticket validation and complaint management system

Out-of-Scope Features:

- Mobile application development (web-based only)
- Social media integration or chat systems
- Advanced analytics and machine learning recommendations
- Video streaming or content hosting
- External shipping/delivery API integration
- Add to Favorite or review system

2. Quality Function Deployment

Quality Function Deployment (QFD) is a method of quality management that converts the needs of customers into technical requirements for the software. The focus of QFD is to maximize customer satisfaction. To achieve this, QFD emphasizes understanding what the customers consider "valuable" and then incorporating those values throughout the engineering process.

QFD defines three types of requirements:

I. Normal Features

II. Expected Features

III. Exciting Features

2.1 Normal Features:

- Email-based registration
- Password login
- Role Selection
- View upcoming concerts list
- See event basic details (date, singer)
- Simple "Buy Ticket" button
- Basic account approval/rejection for only Organizer and Singer.
- Simple complaint viewing

2.2 Expected Features:

- OTP email verification (6-digit, 5-minute expiry)
- Password recovery with OTP
- Password strength requirements (8-12 chars, mixed case, digit, special char)
- Account status checking (active/pending/suspended)
- Edit profile information
- Change profile picture
- Singer: View incoming concert requests

- Organizer: View event status
- Singer availability calendar
- Fixed, non-negotiable performance fees
- Booking request system (venue + city only)
- One-click accept/reject by singers
- Payment integration for performance fees
- Ticket cart (max 10 tickets/event/user)
- Multiple ticket tiers with pricing
- QR ticket generation
- Email delivery of tickets
- Complain submission with audio,texts and videos
- Shopping cart
- Checkout with shipping address collection providing location
- Manual account approval/rejection
- User suspension capability
- Complaint review system

2.3 Exciting Features:

- Dynamic ticket pricing
- Custom event sub-domains (arusha.gaanbajna.com)

3.User Story:

The platform welcomes new users, who can register by choosing one of three roles: Audience, Singer, or Organizer. During registration, everyone must provide a unique username, a valid email, and a strong password. Singers and Organizers are additionally required to upload a profile picture. Upon submitting their details, the system checks if the email is unique. If it is, a 6-digit OTP is immediately sent to that email, and the user is redirected to a verification page with a strict 5-minute time limit. If the correct OTP is entered in time, the email is verified. For Audience members, this completes registration, and they can log in immediately. However, for Singers and Organizers, the account is placed in a "pending" status, triggering a notification to the admin panel.

The admin then contacts the pending user via email to schedule an in-person meeting for identity verification. Once the admin manually approves the account from the panel, the system sends a confirmation email, activating the account. If the OTP is incorrect or expires, the user must request a new one or restart. Notably, Admin accounts are not created publicly; they are generated internally when an existing admin sends a one-time invitation link via email.

Once registered, users can log in using their email or username and password. If the credentials belong to an Audience member or an already-approved Organizer or Admin, they are logged in immediately and redirected to their role-specific homepage. However, if a Singer or Organizer attempts to log in while their account is still pending, they see a message stating, "Your account is under review. We will contact you soon," and are denied access. All users can be logged into multiple devices at once.

Logged-in users can manage their profiles and security. Anyone can view and update their profile picture, username, and email. Singers and Audience members can also manage a saved mobile banking number. For security, a forgotten password can be reset via a time-sensitive email OTP, or changed after logging in by confirming the current password. All new passwords must meet the platform's strength requirements, and every successful change is confirmed via email.

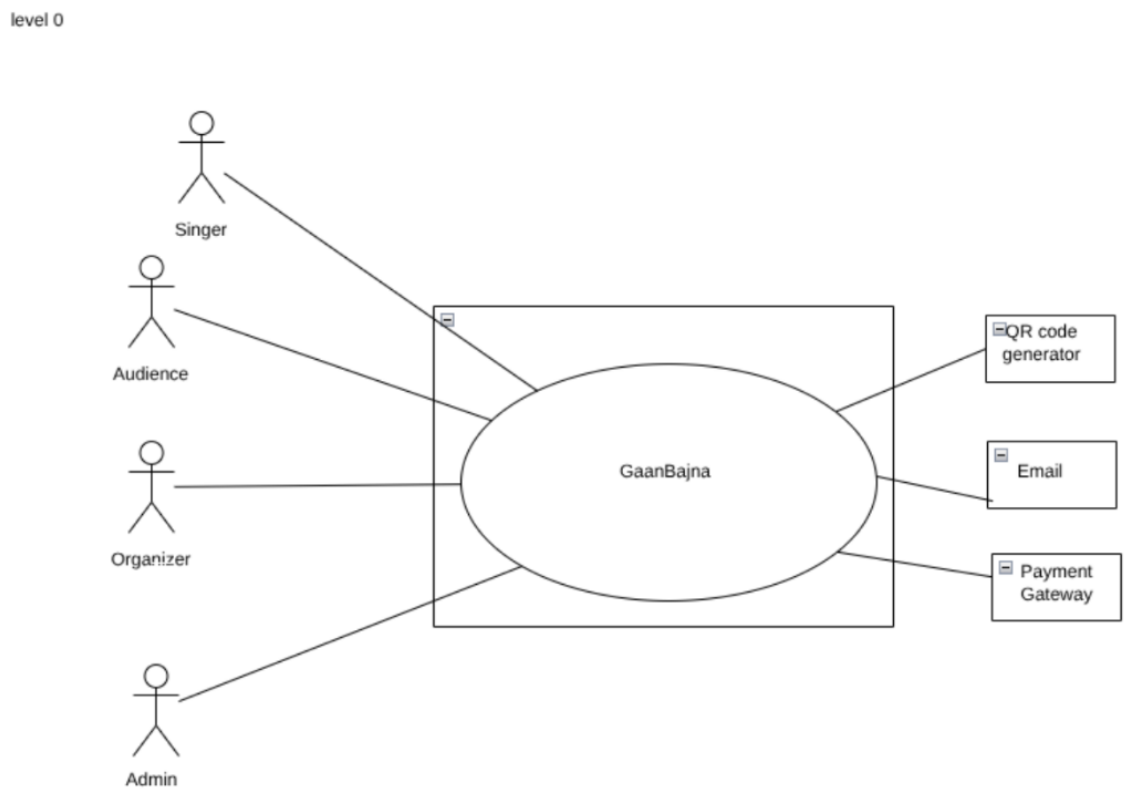
A key function is enabling Organizers to book Singers for events. Organizers can browse Singer profiles, see their availability and fixed fees, and request a slot by selecting a date and providing a venue and city. The Singer is instantly notified and can accept or reject the request without negotiation. If accepted, the Organizer must pay the fee upfront. After payment, the Organizer can create the concert by adding a title, setting up to three ticket prices, uploading a poster, and optionally enabling dynamic pricing. The event then appears on the Singer's profile. Before launch, the Organizer may also purchase a custom web address, which requires admin approval and an extra payment. Once approved, the custom address is active until the concert ends. The concert then goes live for ticket sales immediately.

Audience members can browse these live concerts, add tickets to their cart with a limit of 10 per event, and pay for them all at once. Upon successful payment, each ticket with a unique QR code is emailed to them and saved in their profile. At the event, organizers scan these QR codes in real time. Valid tickets are marked as used instantly, while invalid or duplicate ones are rejected on the spot. If an audience member has an issue, they can submit a complaint with photo or video proof through the concert page or ticket email, which is sent instantly to both the event organizer and the admin.

Beyond events, the platform features a Marketplace. Authorized Singers and Organizers can add merchandise stock with details like price, quantity, description, and photos, and can tag connected events. Audience members cannot add stock. All logged-in users can browse the Marketplace or concert pages to view available merchandise. A personalized "Recommended for You" section appears based on their browsing history. Users can add items to their cart and proceed to checkout, where they must provide mandatory shipping details for every order. After reviewing the summary, they pay the total in one transaction. Following successful payment, the seller arranges for the courier to ship the products.

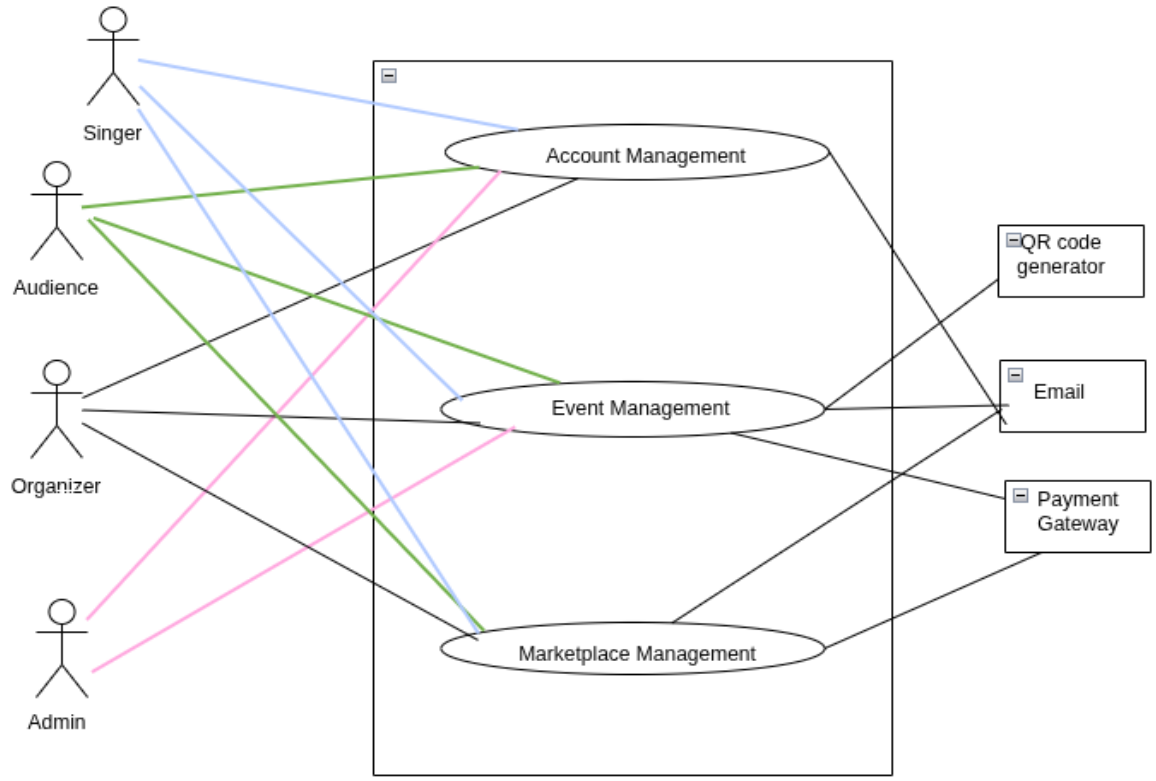
4.Use Case Diagram:

Level 0: GaanBajna
Primary Actors: Admin,Singer,Organizer,Audience
Secondary Actros: QR code generator, Payment Gateway, Email



Level 1: GaanBajna
Primary Actors: Admin,Singer,Organizer,Audience
Secondary Actros: QR code generator, Payment Gateway, Email

level 1



Description:

Account Management

- Manages the **full lifecycle** of a user's identity on the platform.
- Serves as the **gateway** for all user roles.
- Ensures **secure, verified, and role-appropriate access**.
- Uses a **structured process** combining automated checks and administrative review.

Event Management

- Manages the **end-to-end process** of creating, selling tickets for, and executing live concert events.
- Operates as a **structured marketplace** connecting Organizers, verified Singers, and Audience members.
- Organizer browses Singer profiles (availability, fixed fee) and sends a booking request (date, venue, city).

- Singer receives instant notification and can **accept or reject** the request with one click (no negotiation).
- Upon acceptance, Organizer must **pay the full performance fee upfront**.
- After payment, Organizer creates the concert: sets title, poster, up to 3 ticket prices (optional dynamic pricing).
- Organizer can optionally **buy a custom web address**, requiring admin approval and extra payment.
- Event goes **live immediately** for ticket sales.
- Audience browses live events, adds tickets to cart (max 10 per event), and pays in one transaction.
- System generates **unique QR-coded tickets**, emails them, and saves them in the user's profile.
- At the venue, Organizer scans QR codes for **real-time validation**.
- Valid tickets are marked "used"; invalid/duplicate tickets are rejected on the spot.
- Audience can submit **complaints with evidence** (photos/videos) via the concert page or ticket email.
- Complaints are sent **instantly to both the Organizer and the Admin**.
- Admin can suspend/ban accounts based on multiple valid complaints, canceling events and notifying the user.

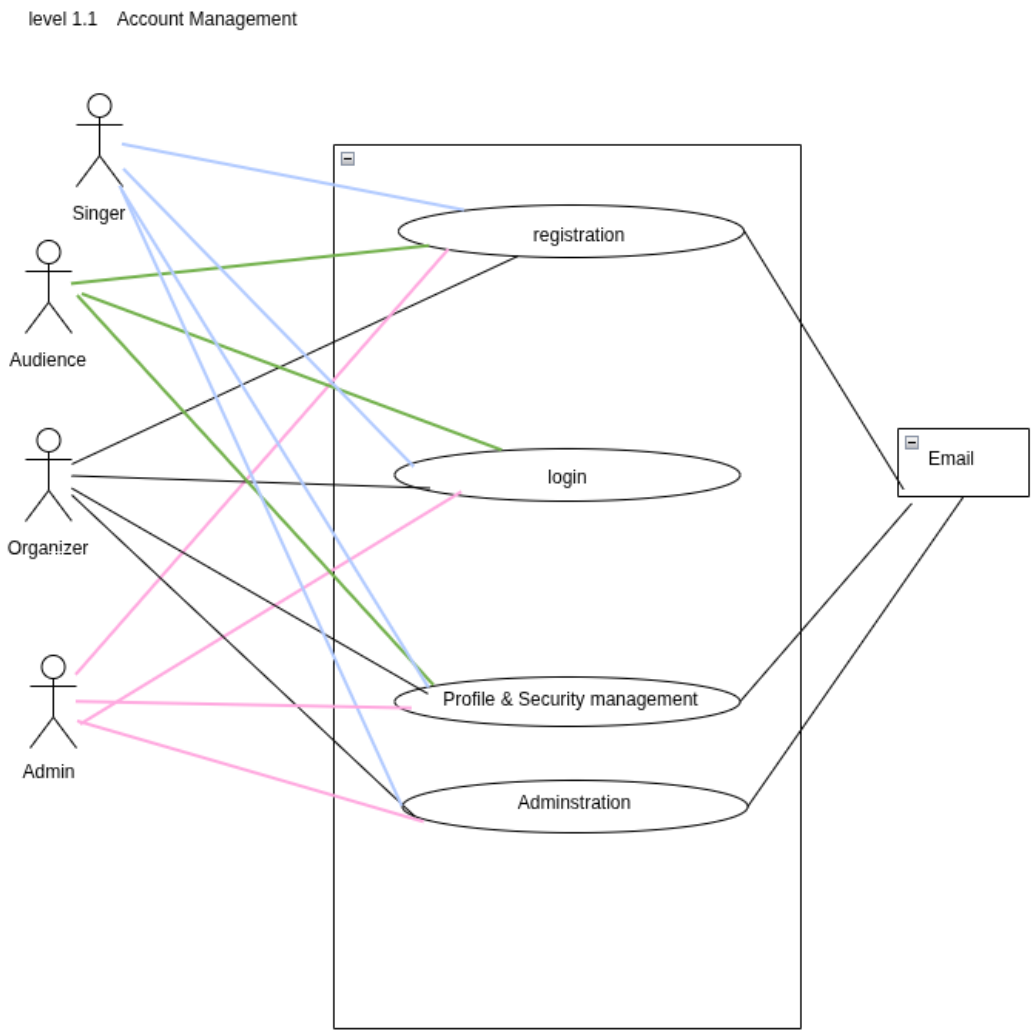
Marketplace Management

- Enables the **buying and selling of physical merchandise** related to concert events.
- **Authorized Sellers:** Only Singers and Organizers can list products.
- **Listing Process:** Sellers specify price, quantity, description, photos, and can tag items to their events.
- **Browsing:** All logged-in users can browse available merchandise.
- **Personalization:** Platform shows a **"Recommended for You"** section based on user history.
- **Checkout Process:**
    - User adds items to cart.
    - At checkout, providing **full shipping details (name, phone, address) is mandatory**.
    - User pays the total amount in **one consolidated transaction**.
- **Fulfillment:** After payment, the responsible seller ships the order to the provided address via courier.

Level 1.1 : Account Management

Primary Actors: Singer,Organizer,Admin,Audience

Scondary Actors: Email



Description:

Registration

- User selects a role: Audience, Singer, or Organizer.
- All users provide a unique username, valid email, and a strong password.
- Singers and Organizers must also upload a profile picture.
- System sends a 6-digit OTP to the email for verification.
- After successful OTP verification:

    - Audience accounts are activated immediately.
    - Singer/Organizer accounts are set to **"pending"** for admin review.
- Admin accounts are created only through an internal invitation from an existing admin.

Login

- Users log in using their registered email/username and password.
- System authenticates credentials and checks account status.
- Access is **granted** to: Audience members, approved Organizers, and Admins.
- Access is **blocked** for Singers/Organizers with "pending" status; a friendly "account under review" message is shown.
- All users can be logged in on multiple devices at the same time.

Profile & Security Management:

- Logged-in users can view and update profile details (username, profile picture).
- Singers and Audience members can also manage a saved mobile banking number.
- A forgotten password is reset via a time-sensitive email OTP.
- Users can change their password after verifying the current one; new passwords must meet strong password requirements.
- The system sends a confirmation email for every successful password change.

Administration

- Admin reviews all **pending** Singer and Organizer accounts.
- Admin contacts user, schedules an in-person meeting, verifies identity/documents, and collects mobile banking details.
- Admin manually **approves** or **rejects** the account from the admin panel.
- System automatically sends a status notification email to the user.
- For disciplinary action, admin can **suspend** accounts with multiple valid complaints.
- Suspension user access, cancels/hides their events, and notifies them via email with the reason.

Action & Reply:

Action: User selects registration role (Audience/Singer/Organizer)
Reply: System displays appropriate registration form with required fields for the selected role.

Action: User submits registration form with unique username, email, and password (Singer/Organizer uploads profile picture)
Reply: System validates password strength and checks if email is unique. If valid, system generates and sends a 6-digit OTP to the user's email.

Action: User enters the received OTP on the verification page.
Reply: System checks OTP correctness and the 5-minute time window. If valid, it verifies the email, activates Audience accounts immediately, and places Singer/Organizer accounts in "pending" status (notifying the admin).

Action: Admin contacts pending user, schedules meeting, verifies identity/documents in-person, and collects mobile banking details.
Reply: Admin manually selects 'Approve' or 'Reject' in the admin panel. System updates the account status accordingly and sends a notification email to the user.

Action: User enters login credentials (username/email and password).
Reply: System authenticates the credentials. If correct and account is active, it grants access. If the account is pending (Singer/Organizer), it displays a "Your account is under review" message and denies login.

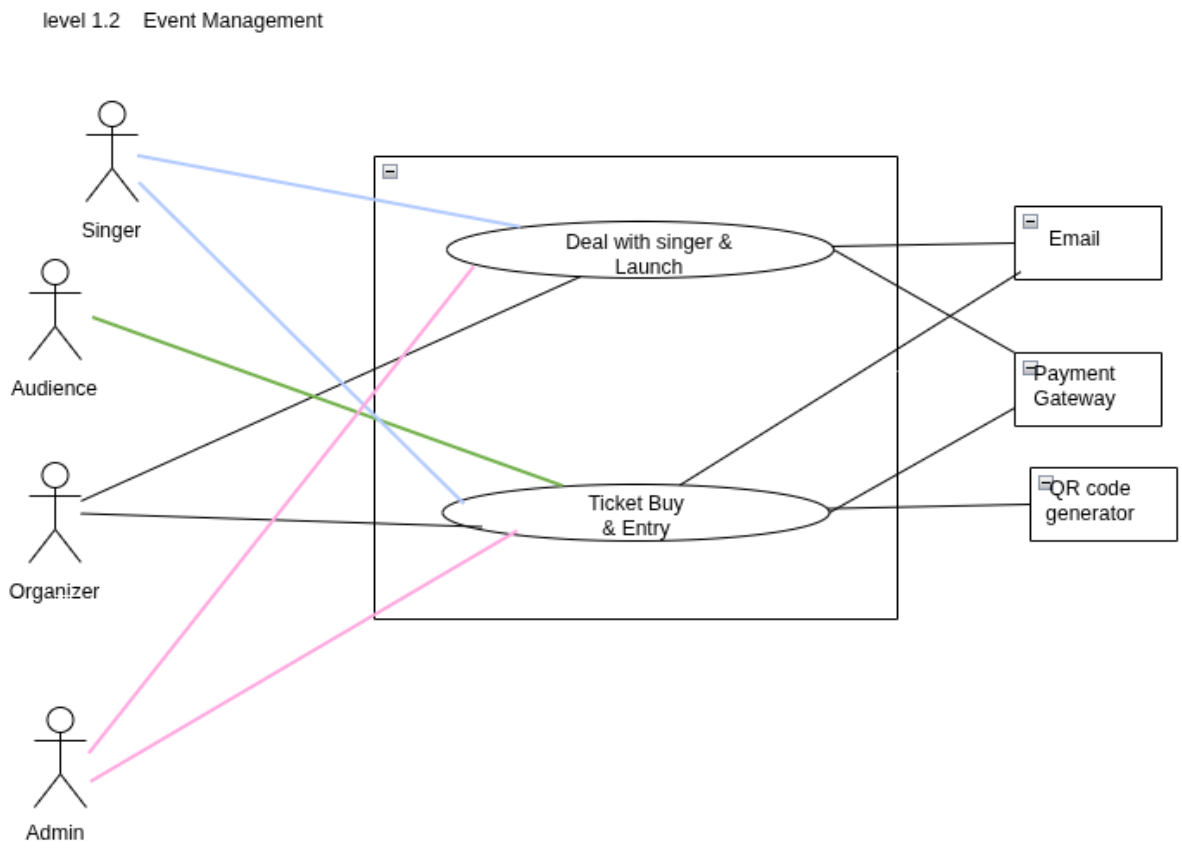
Action: A logged-in user requests a password reset via the 'Forgot Password' link.
Reply: System sends a time-sensitive OTP to the user's registered email for verification.

Action: A logged-in user updates their profile information (e.g., profile picture, mobile banking number) or changes their password after verifying the current one.
Reply: System validates and saves the new information. For a password change, it sends a confirmation email to the user.

Level 1.2: Event Management

Primary Actors: Admin,Singer,Organizer,Audience

Secondary Actros: QR code generator, Payment Gateway, Email



Description:

Deal with Singer & Launch

- Organizer Browse verified Singer profiles to see availability and fixed fees to select a performer.
- Organizer Book a Singer by selecting a date and entering only venue/city, sending a binding, non-negotiable request.
- Singer Receive instant notification of booking requests.
- Singer Review request details and accept or reject it with a single, non-negotiable click.
- Organizer Upon Singer's acceptance, pay the full performance fee upfront.
- Organizer Create the concert after payment.
- Organizer Set event title, poster, up to 3 ticket prices (option for dynamic pricing).
- Organizer Purchase a custom web address for the event (requires extra payment and admin approval).

- System Launch the event immediately after setup, listing it on the Singer's profile and opening it for public ticket sales.

Ticket Buy & Entry

- Audience Browse active concerts and purchase tickets.
- Audience Add tickets to cart (max 10 per event), review cart, and pay for all tickets in one transaction.
- System After payment, generate a unique QR code per ticket, email tickets, and store them in the user's profile.
- Organizer Validate attendee entry at the venue.
- Organizer Scan each attendee's ticket QR code in real-time.
- System Instantly validate code. Mark valid tickets as "used" and reject invalid/duplicate tickets on the spot.
- Audience Submit a complaint about an event.
- Audience File complaint via concert page or ticket email, including description and photo/video evidence.
- System Route complaint instantly and simultaneously to the Event Organizer and the Admin.
- Admin Take disciplinary action based on complaints.
- Admin Suspend or ban an Organizer/Singer account after multiple valid complaints.
- System Immediately block user access, cancel/hide their events, and send a notification email stating the reason.

Action & Replies

Action: Organizer browses verified Singer profiles.
Reply: System displays Singer list with their available dates and fixed performance fees.

Action: Organizer selects a Singer, chooses a date, and enters venue/city to send a booking request.
Reply: System sends an instant, binding booking notification to the selected Singer.

Action: Singer receives and reviews the booking request details.
Reply: System presents request summary with "Accept" and "Reject" buttons.

Action: Singer clicks 'Accept' or 'Reject'.

Reply: System notifies the Organizer of the decision. If accepted, the system prompts the Organizer for payment.

Action: Organizer pays the Singer's full performance fee.
Reply: System confirms payment and unlocks the concert creation form for the Organizer.
Action: Organizer creates the concert by setting the title, poster, and up to 3 ticket prices (optionally enabling dynamic pricing).
Reply: System saves the concert draft. If a custom web address is purchased, it routes the request for admin approval.

Action: Organizer finalizes and launches the concert.
Reply: System publishes the event immediately, lists it on the Singer's profile, and opens it for public ticket sales.

Action: Audience browses active concerts and adds tickets to their cart (max 10 per event).
Reply: System updates the cart and enforces the ticket limit.

Action: Audience proceeds to checkout and pays for all cart tickets in one transaction.
Reply: Upon successful payment, the system generates a unique QR code for each ticket, emails them to the user, and saves them in the user's profile.

Action: Organizer (or staff) scans an attendee's ticket QR code at the venue.
Reply: System validates the code in real-time, marks valid tickets as "used," or instantly rejects invalid/duplicate tickets.

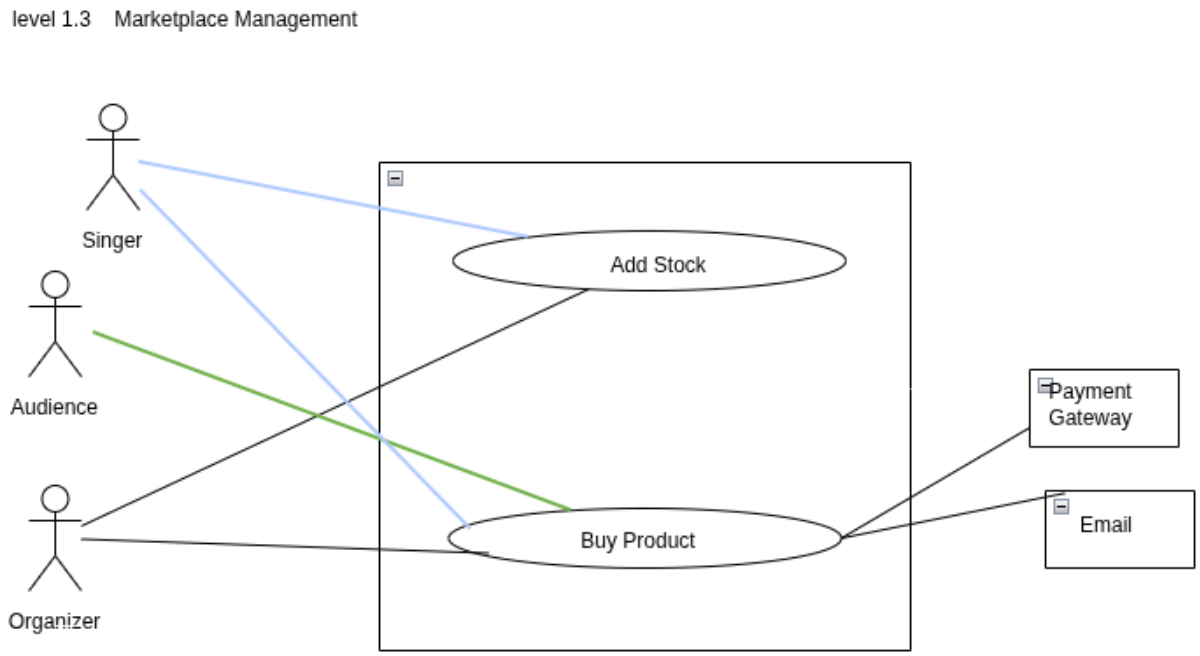
Action: Audience files a complaint via the concert page or ticket email, including description and evidence.
Reply: System instantly routes the complaint to both the Event Organizer and the Admin.

Action: Admin reviews multiple valid complaints against a user and chooses to suspend or ban the account.
Reply: System immediately blocks the user's access, cancels/hides their events, and sends them a notification email stating the reason.

Level 1.3: Marketplace Management

Primary Actors: Singer,Organizer,Audience

Secondary Actros: Payment Gateway, Email



Description:

Add Stock:

This module allows authorized users to list merchandise for sale. It is restricted to users with selling privileges—specifically, verified Singers and Organizers. The process begins when a seller decides to add a new product. They must provide essential listing details, including the item's price, available quantity, a clear description, and multiple photos to showcase the product. Additionally, the seller has the option to tag the merchandise to one of their upcoming or past events for targeted promotion. Once submitted, the system validates the information and publishes the item to the marketplace, making it visible to all potential buyers.

Buy Product:

This module facilitates the purchase of merchandise by any logged-in user. Buyers can browse all available stock in the general marketplace or view items tagged to specific concert pages. The platform enhances discovery by displaying a personalized "Recommended for You" section based on the user's browsing and purchase history. When a user finds an item, they can add it to their cart in any available quantity. To complete a purchase, the user proceeds to checkout, where they must provide or confirm mandatory shipping information—full name, phone number, and complete delivery address—for every order. After reviewing the final order summary and total cost, the user completes a single, consolidated payment via the platform's payment gateway. Following a successful transaction, the system confirms the order, and the responsibility shifts to the seller to arrange shipment via courier to the provided address.

Action & Reply

Action: Authorized Seller (Singer/Organizer) adds a new product listing with price, quantity, description, and photos.
Reply: System validates details, saves the listing, and publishes it to the marketplace. Seller can optionally tag it to a connected event.

Action: Any logged-in user browses the marketplace or concert pages.
Reply: System displays all available merchandise with details (photos, description, price, stock) and a personalized "Recommended for You" section.

Action: User adds available merchandise to their cart in desired quantities.
Reply: System updates the cart in real-time, reflecting the selected items and quantities.

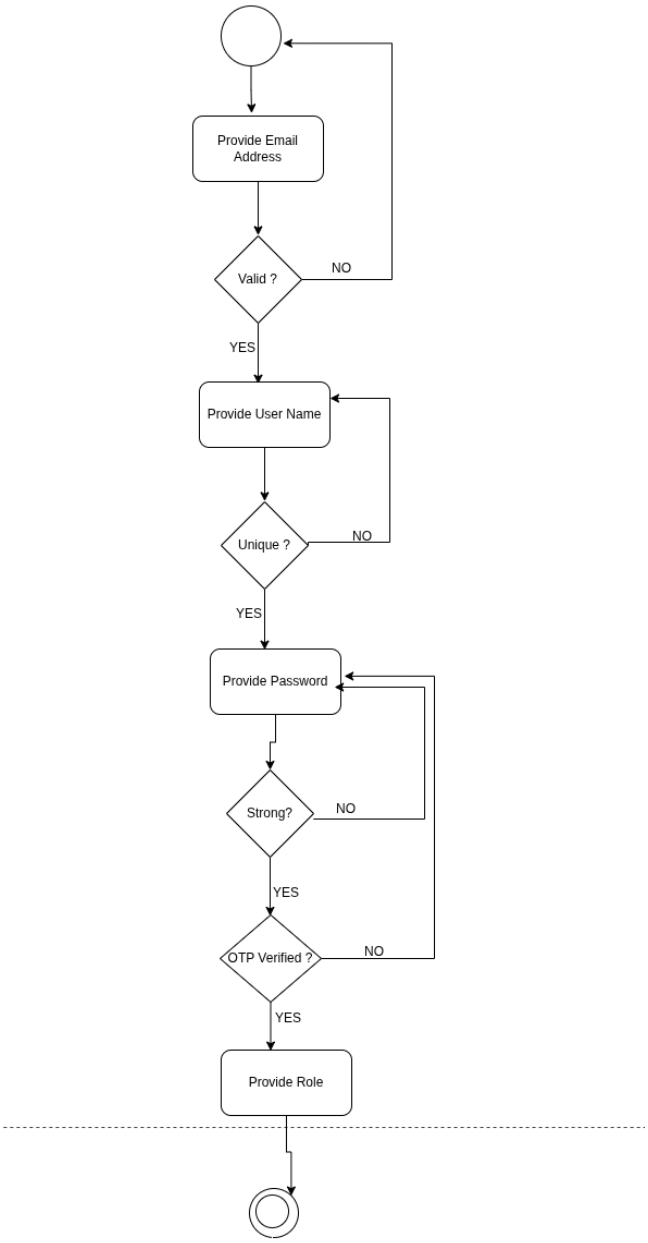
Action: User proceeds to checkout from the cart.
Reply: System prompts the user to fill in or confirm mandatory shipping details (full name, phone number, complete address).

Action: User submits shipping information and reviews the final order summary.
Reply: System displays the order summary with the total amount and requests payment.

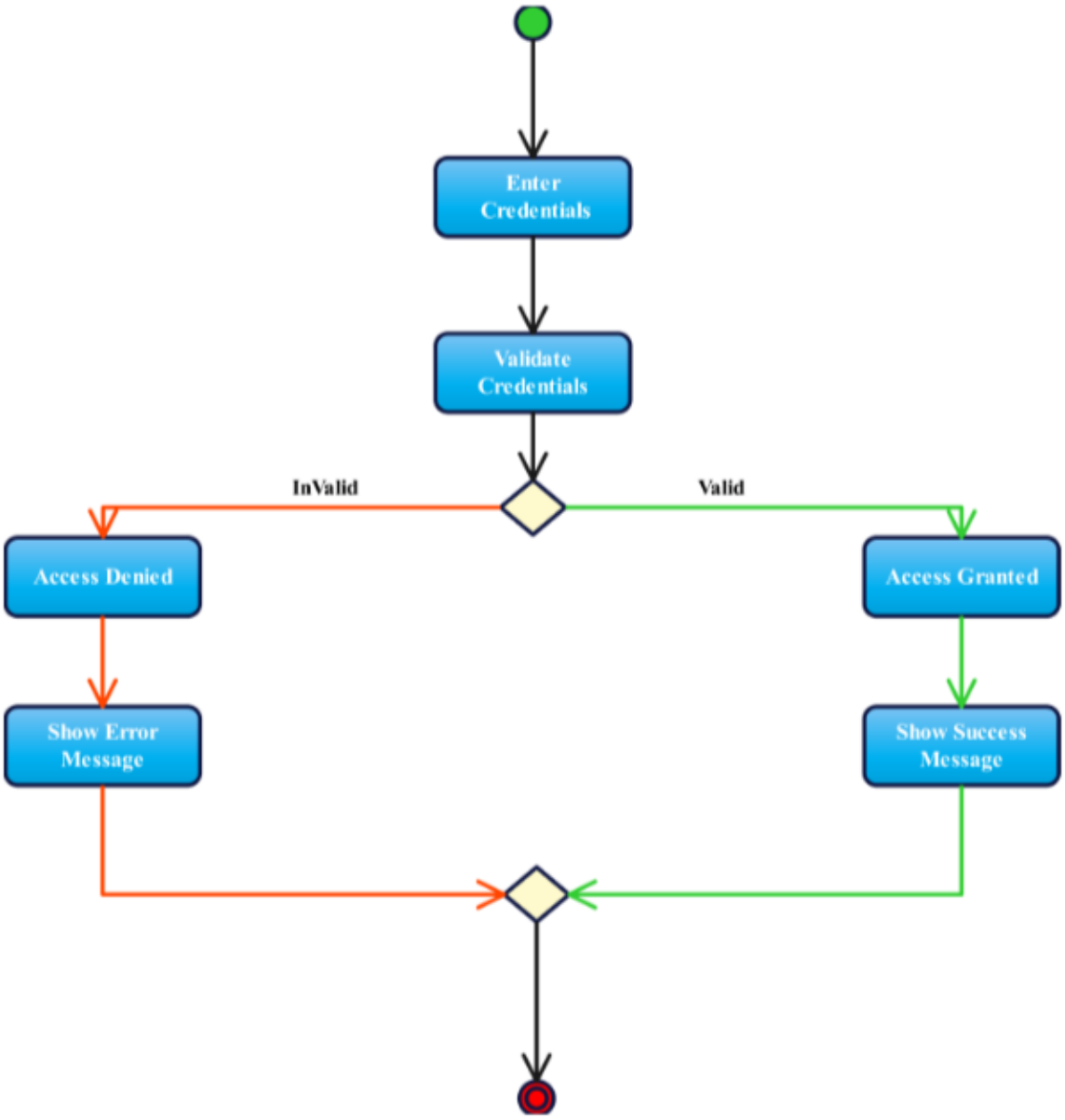
Action: User completes the single, consolidated payment for the entire order.
Reply: System confirms the successful payment, notifies the seller, and marks the purchased stock as reserved. The seller is now responsible for shipment.

5. Activity Diagram:

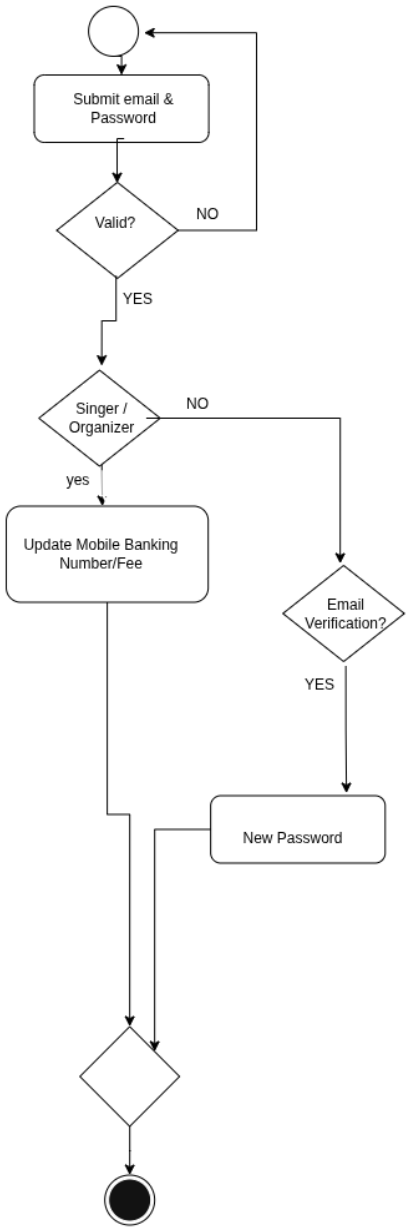
Level 1.1.1 Registration:



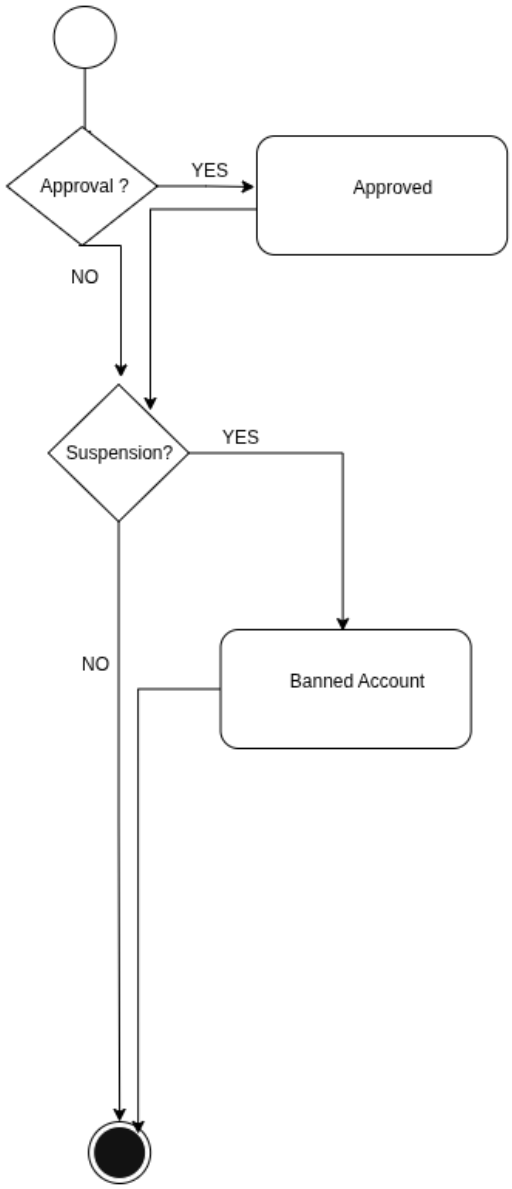
Level 1.1.2 Login:



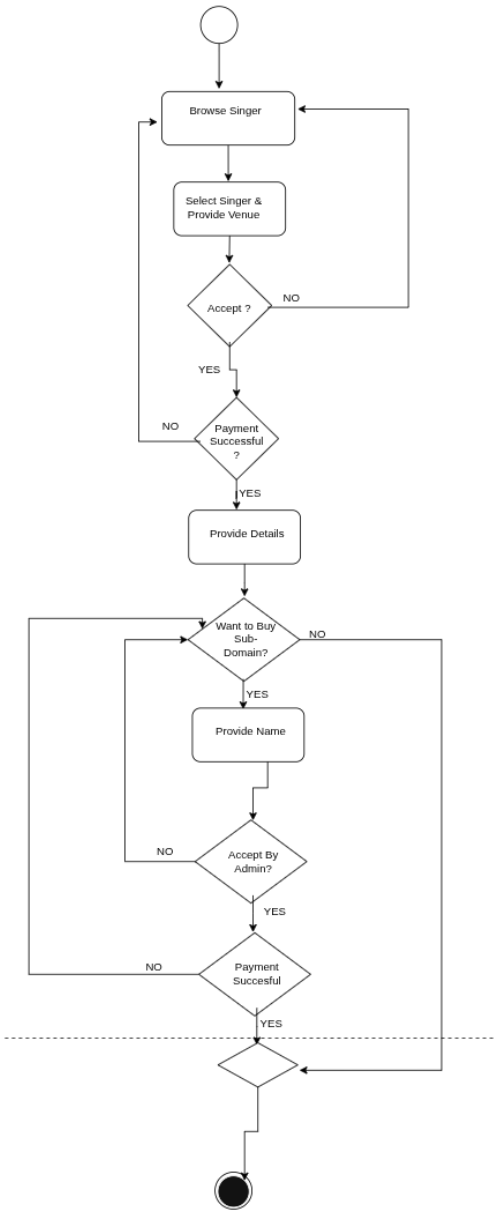
Level 1.1.3 Profile & Security Management:



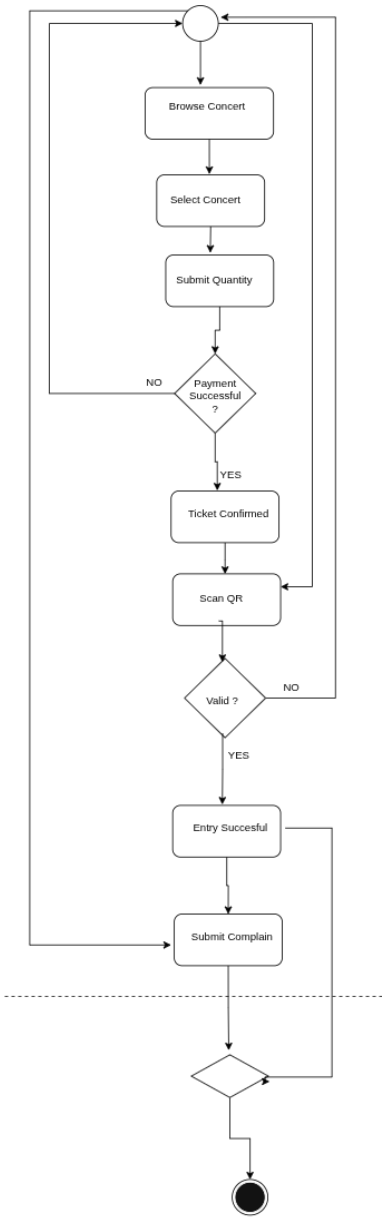
Level1.1.4 Administration



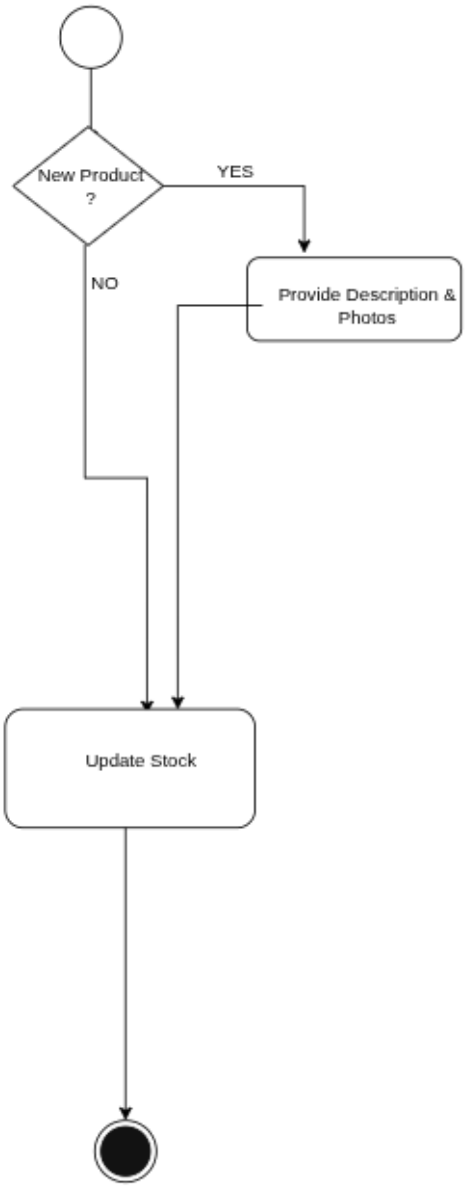
Level 1.2.1 Deal With Singer & Launch:



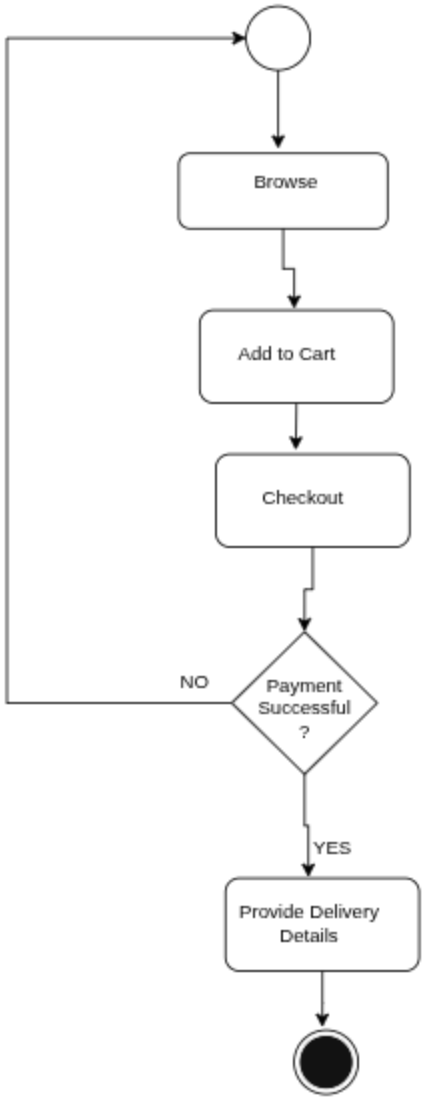
Level 1.2.2 Ticket Buy & Entry



Level 1.3.1 Add Stock

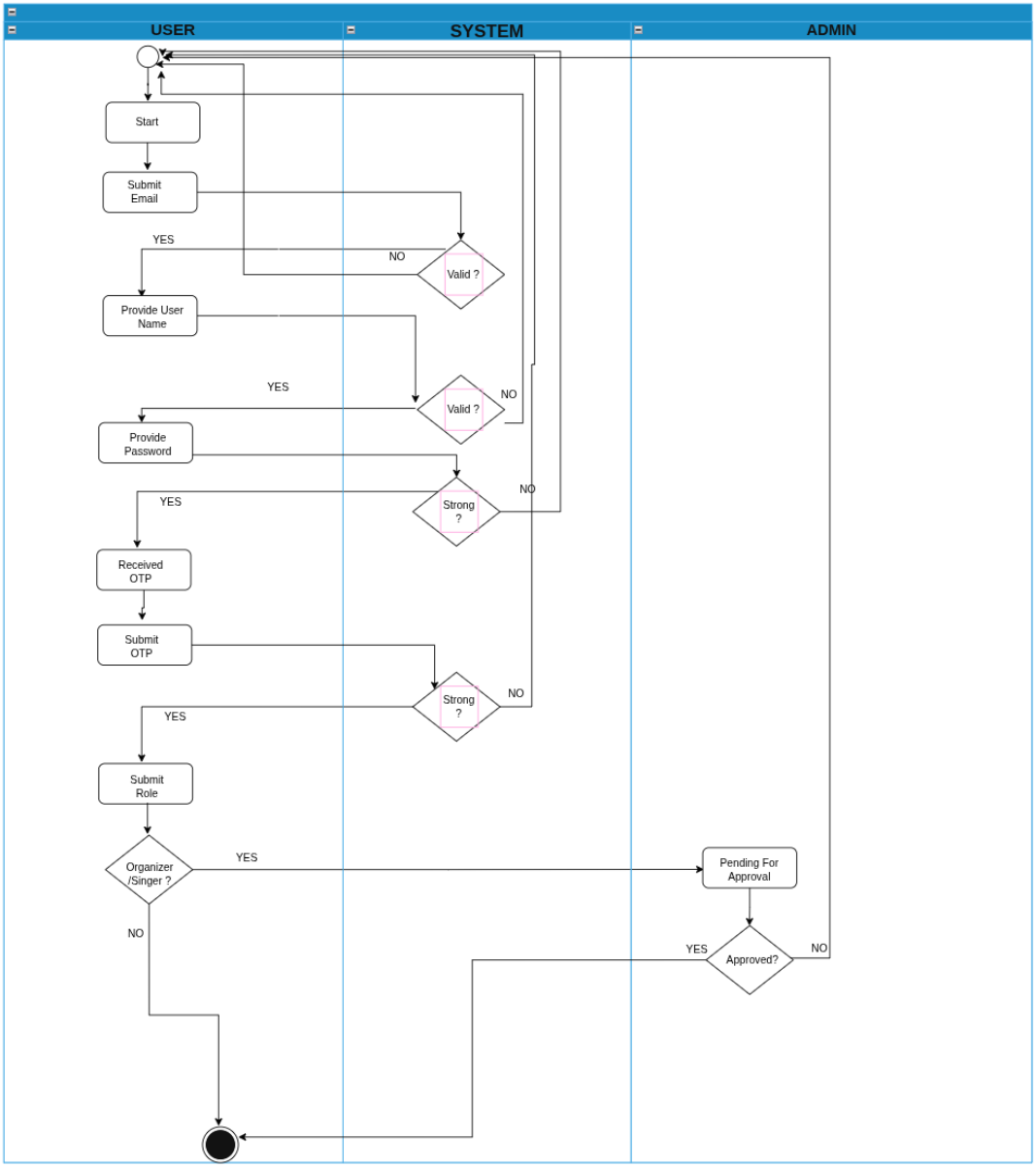


Level 1.3.2 Buy Product

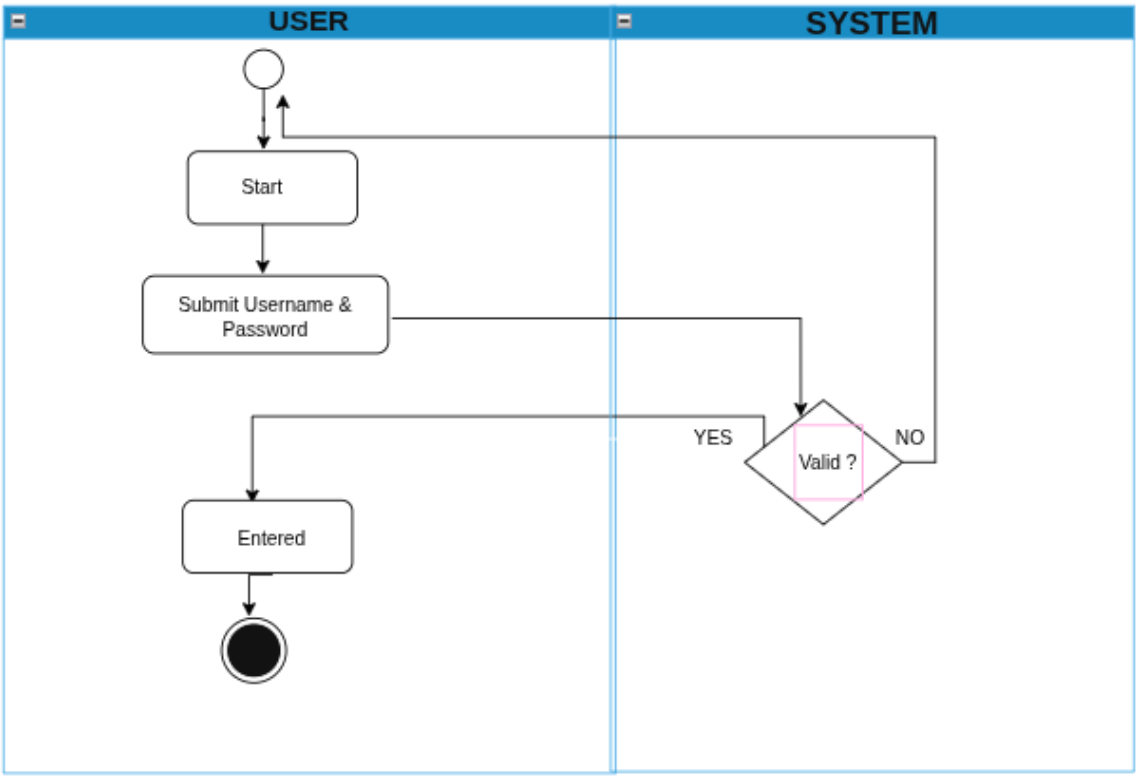


6. Swimlane Diagram:

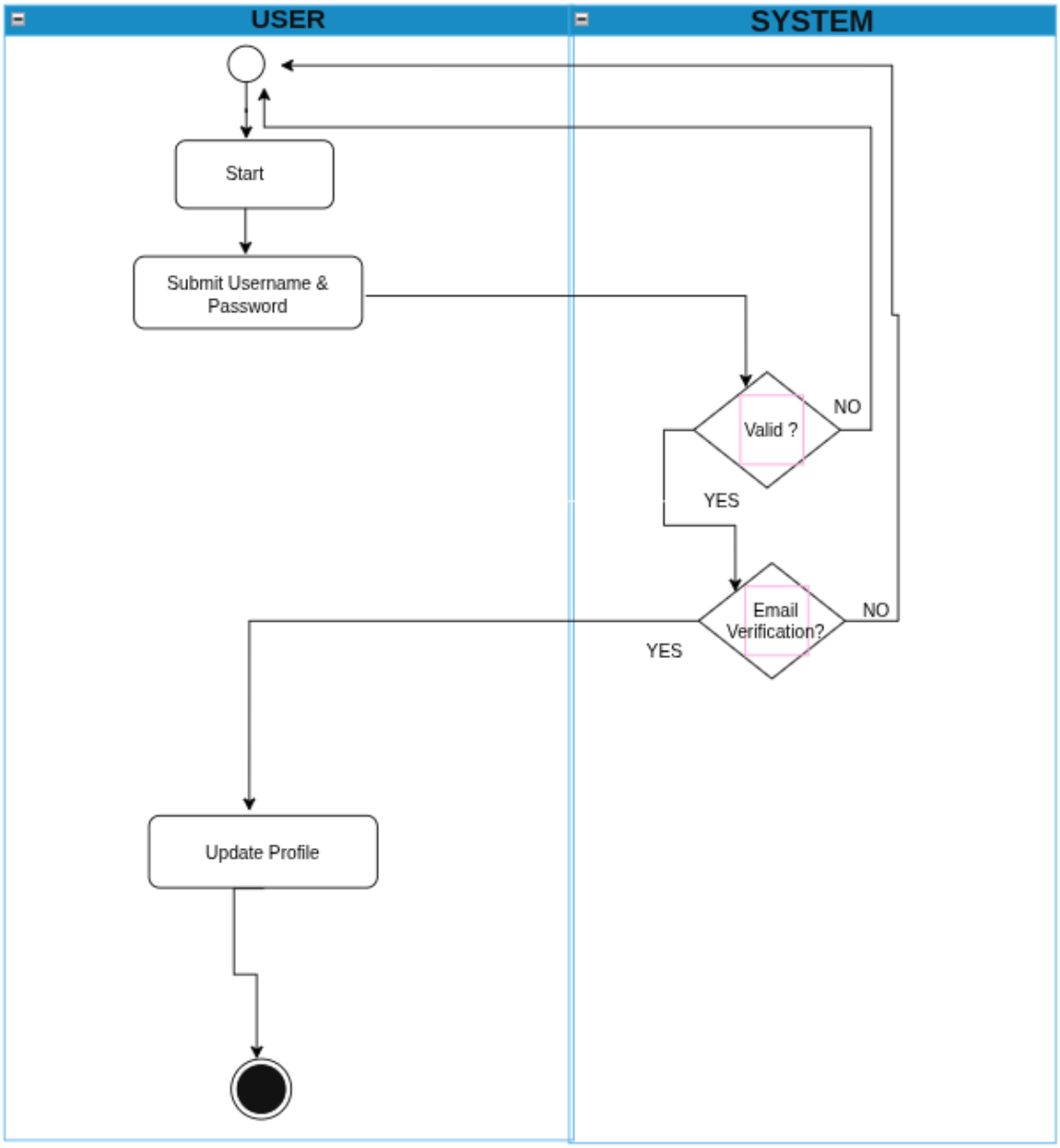
Level 1.1.1 Registration



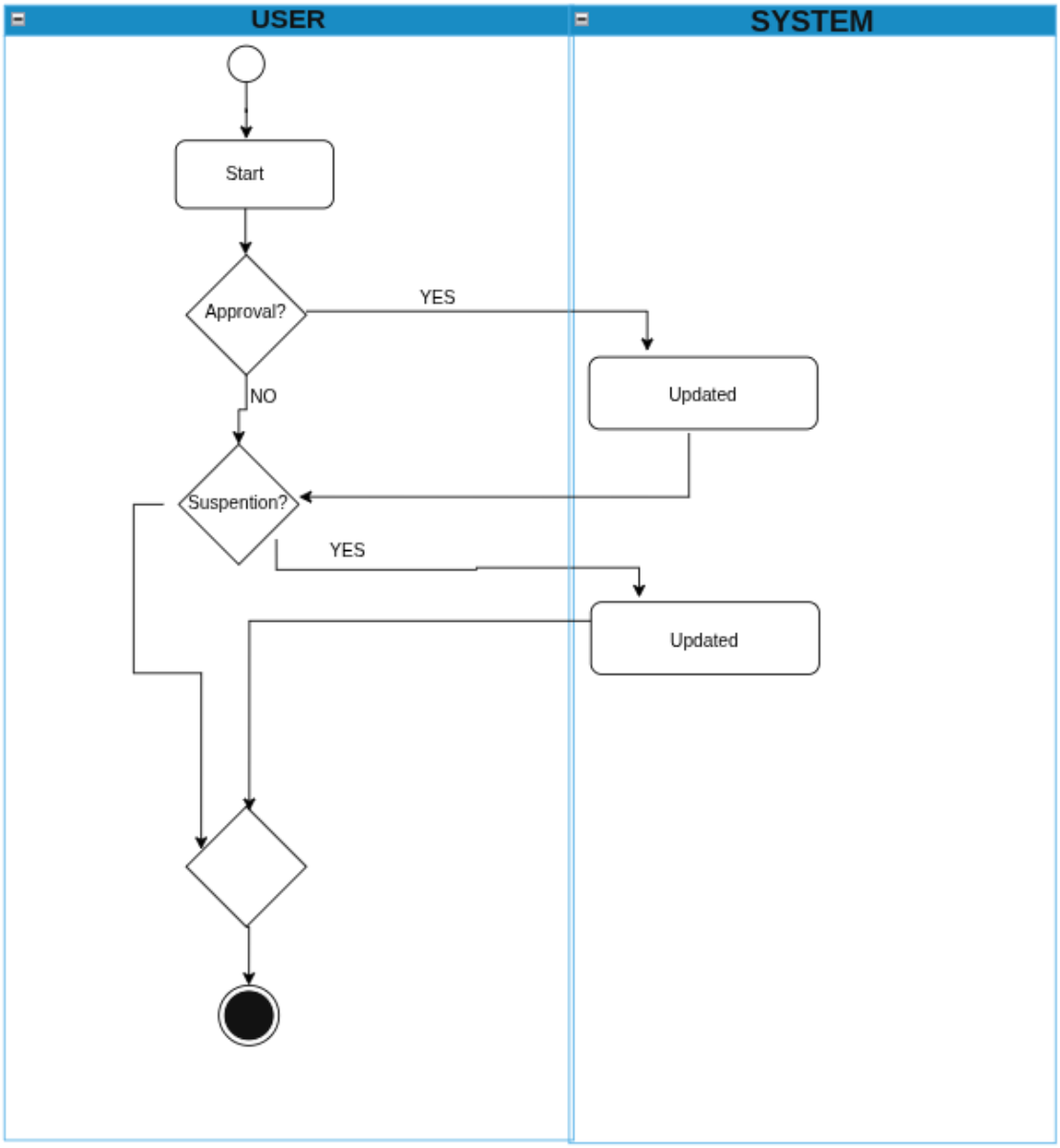
Level 1.1.2 Login



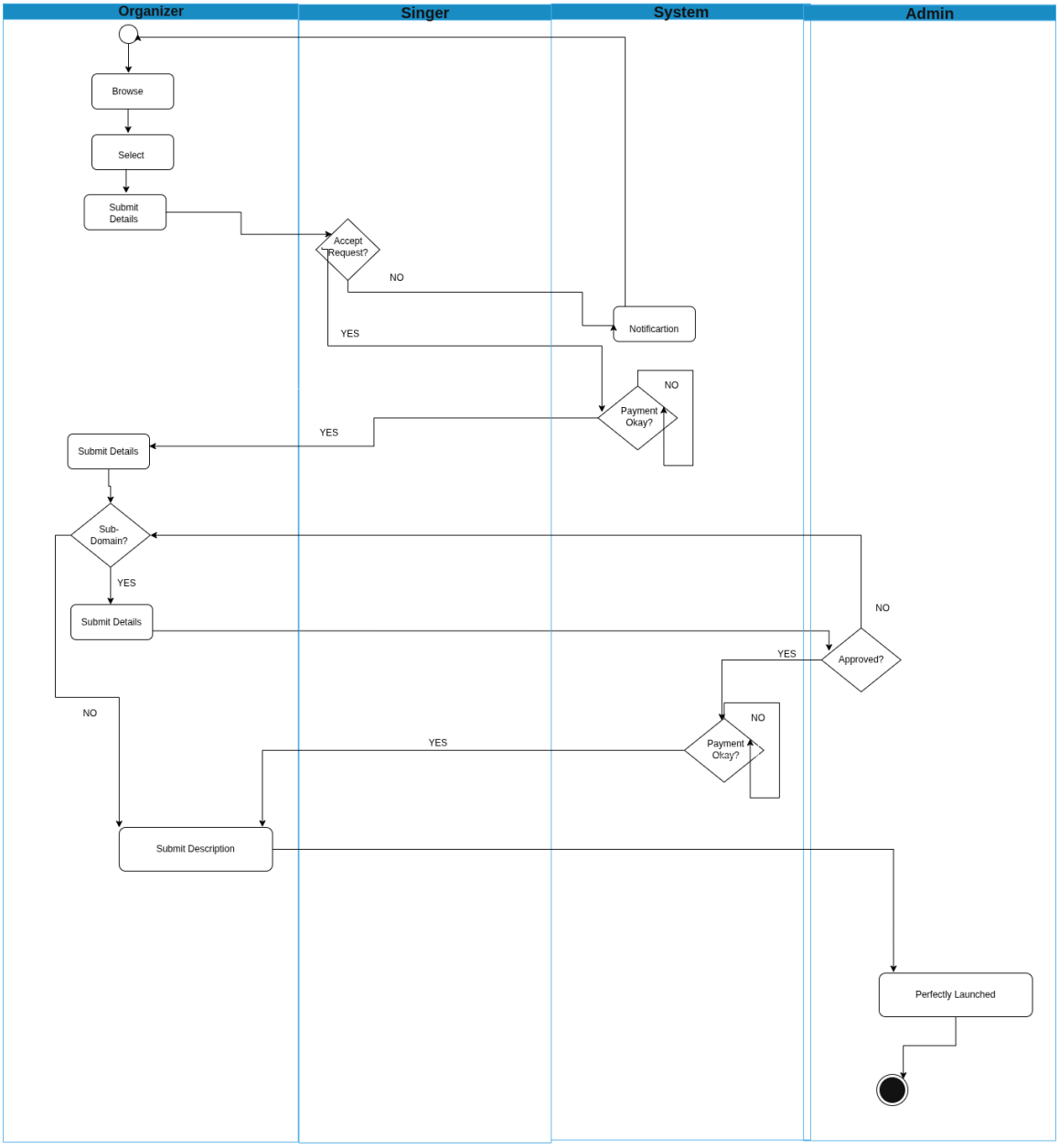
Level 1.1.3 Profile & Security Management



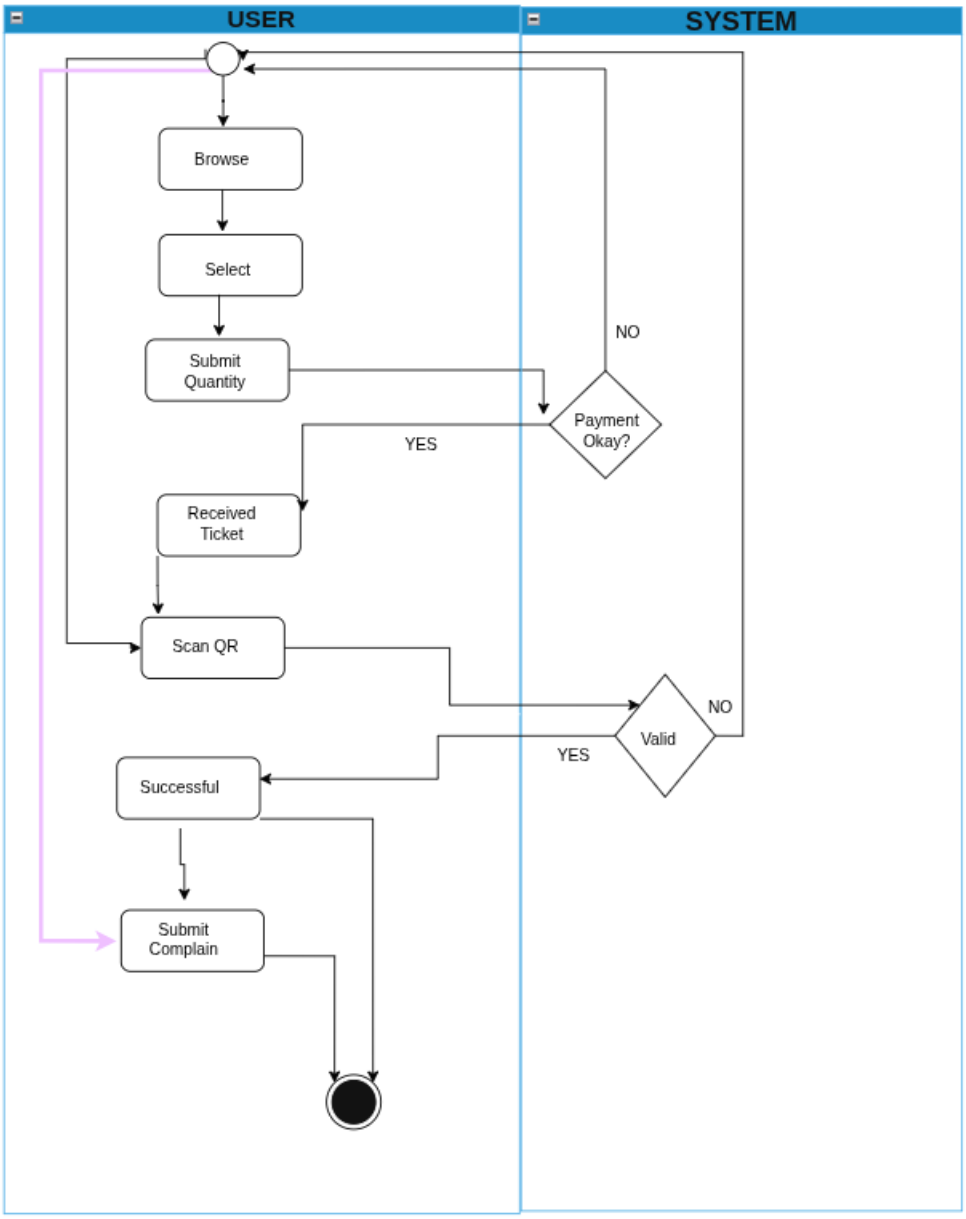
Level 1.1.4 Administration



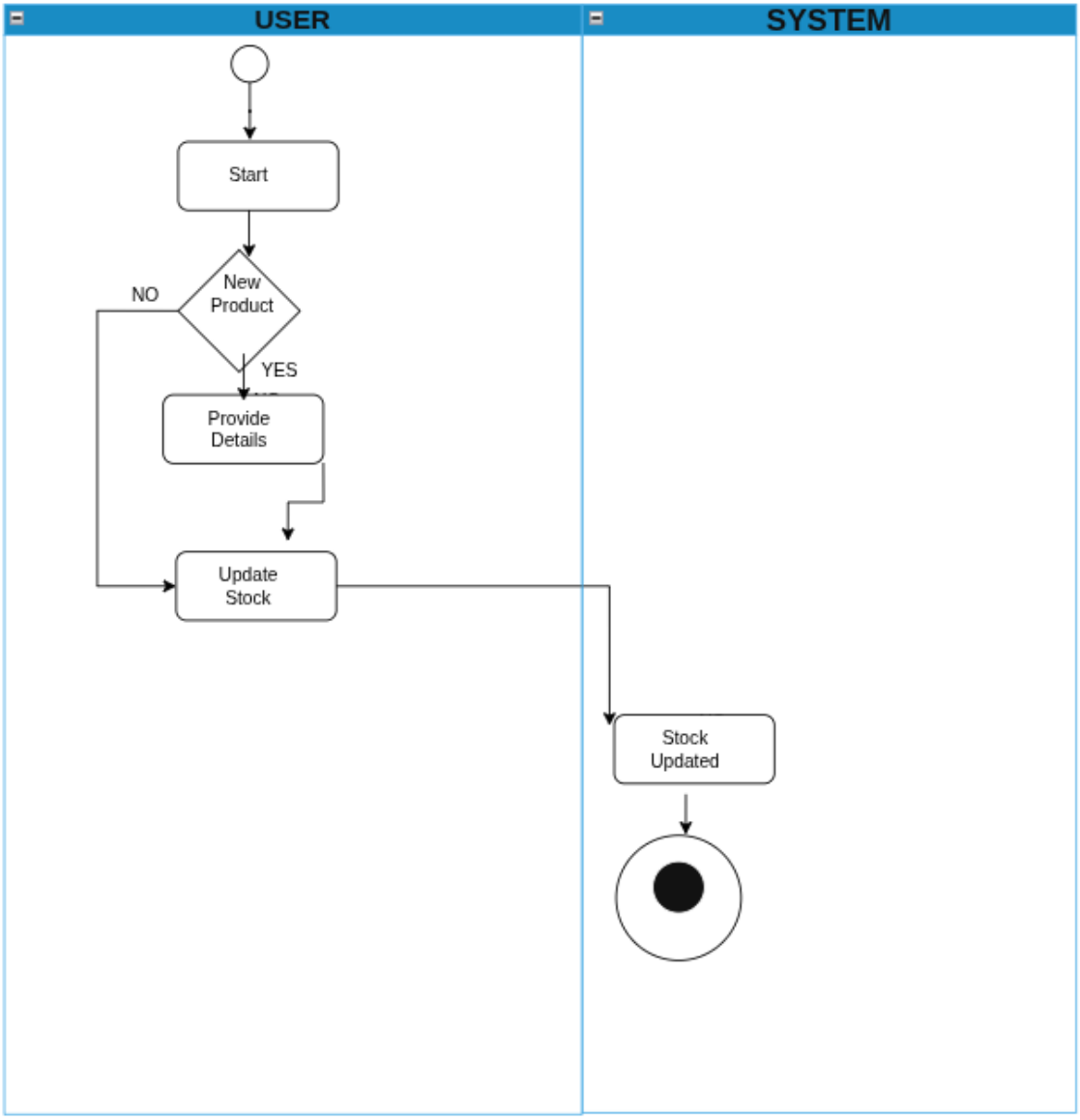
Level 1.2.1 Deal With Singer & Launch



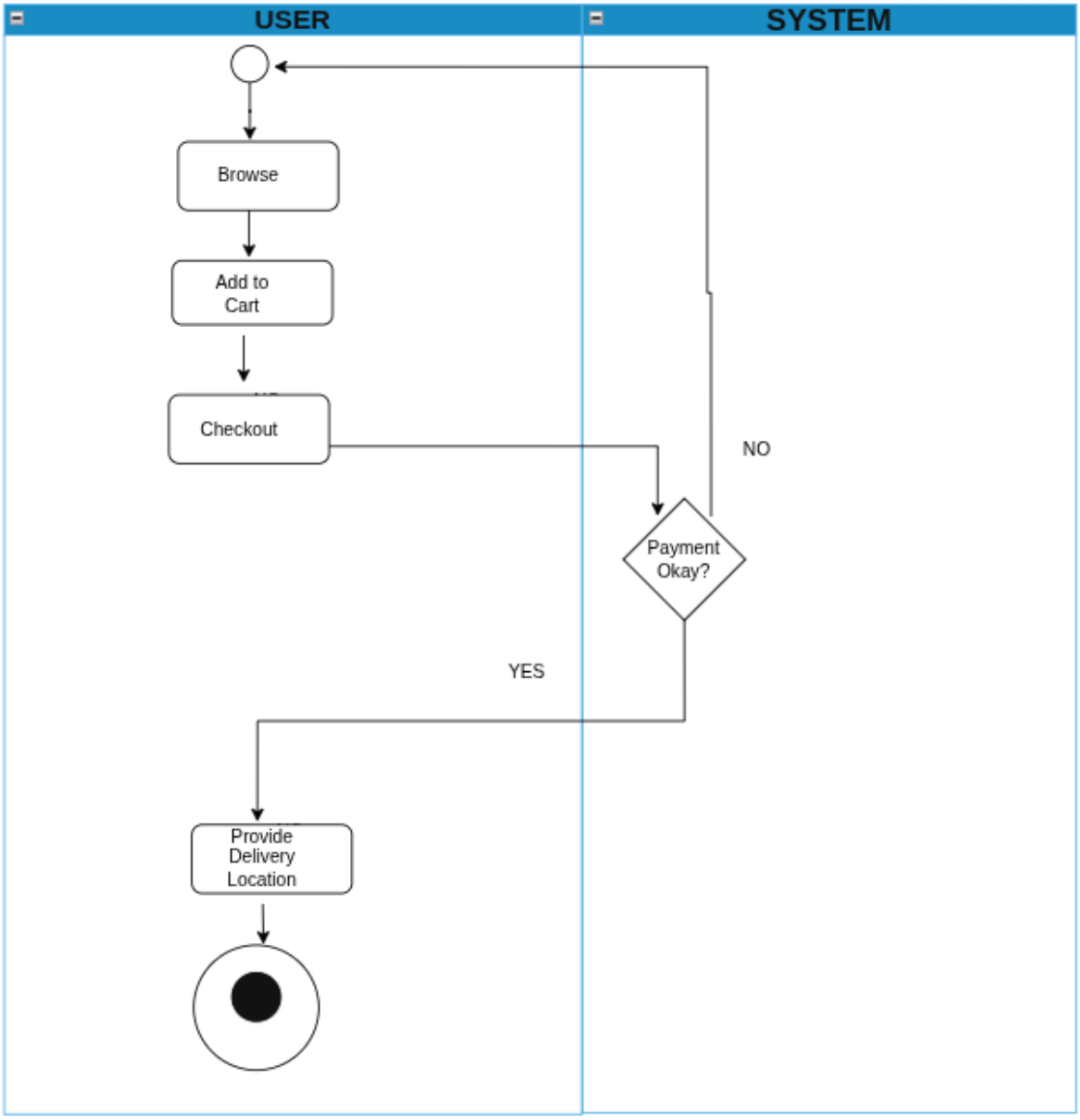
Level 1.2.2 Ticket Buy & Entry



Level 1.3.1 Add Stock



Level 1.3.2 Buy Product


USER
SYSTEM
Browse
Add to Cart
Checkout
NO
Payment Okay?
YES
Provide Delivery Location

7. Data Based Modeling:

If software requirements include the necessity to create, extend or interact with a database or complex data structures need to be constructed and manipulated, then the software team chooses to create data models as part of overall requirements modeling. The entity relationship diagram (ERD) defines all data objects that are processed within the system, the relationships between the data objects and the information about how the data objects are entered, stored, transformed and produced within the system.

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Data Object Identification:

Serial	Nouns	P/S	Attributes
1.	User	**P**	
2.	Email Address	**P**	
3.	Username	**P**	
4.	Password	**P**	
5.	Profile Picture	**P**	
6.	Role	**P**	
7.	Status	**S**	1,17,23,30,31,33,34,35,47,11
8.	Mobile Banking Number	**P**	
9.	Shipping Address	**P**	
10.	Calendar	**P**	
11.	Performance fee	**P**	
12.	Audience	**P**	
13.	Singer	**P**	
14.	Organizer	**P**	
15.	Admin	**P**	
16.	Notification	**S**	1,11,19,30,32,46
17.	Session	**P**	
18.	Device	**P**	

19.	Suspension	**S**	1,52
20.	Profile	**S**	**1,7**
21.	Dashboard	**S**	**20,1,7**
22.	Event	**P**	
23.	Slot	**P**	
24.	Venue	**P**	
25.	City	**P**	
26.	Title	**P**	
27.	Poster	**P**	
28.	Description	**P**	
29.	Dynamic Pricing	**S**	**1**
30.	Custom Sub-Domain	**S**	**1,22,34,33**
31.	Ticket Tier	**P**	
32.	Event Request	**P**	
33.	Request Status	**S**	**1**
34.	Payment	**P**	
35.	Payment Gateway	**S**	**1,34**
36.	Ticket	**P**	
37.	QR code	**S**	**1,36**
38.	Cart	**S**	**1,34**
39.	Checkout	**S**	**1,38**
40.	Marketplace	**P**	
41.	Merchandise	**P**	
42.	Stock	**P**	
43.	Price	**P**	
44.	Quantity	**P**	
45.	Seller	**P**	
46.	Complaint	**P**	
47.	Evidence	**P**	
48.	Audio	**P**	
49.	Video	**P**	
50.	Photos	**P**	
51.	Text	**P**	
52.	Admin Panel	**S**	**1,47,7**
53.	OTP	**S**	**1,54**
54.	Expiry Time	**P**	
55.	Attempt Count	**P**	
56.	Email Verification	**P**	
57.	Password Reset	**P**	
58.	Verification Meeting	**S**	**1,13,14,15**
59.	Meeting Date/Time	**P**	

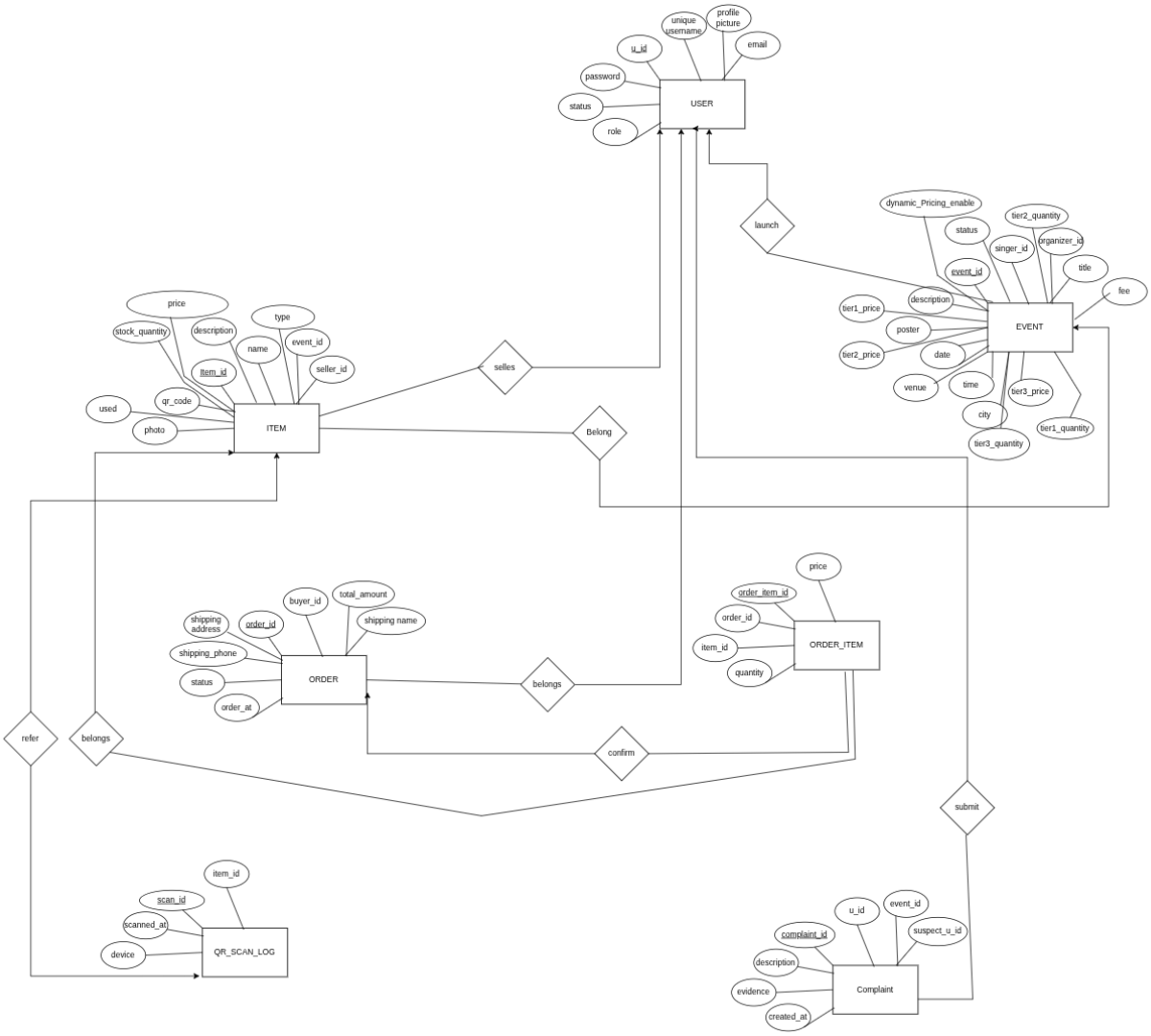
60.	Support Contact	**S**	**1,13,14,15**
61.	Platform	**P**	
62.	In_person meeting	**P**	
63.	credentials	**P**	
64.	access	**P**	
65.	Homepage	**P**	
66.	Messages	**P**	
67.	Devices	**P**	
68.	Negotiation	**P**	
69.	cost	**P**	
70.	Web-address	**P**	
71.	limit	**P**	
72.	QR_code	**P**	
73.	fraud	**P**	
74.	grievance	**P**	
75.	issue	**P**	
76.	Photo Proof	**P**	
77.	Staff	**P**	
78.	Oversight	**P**	
79.	Evidence	**P**	
80.	Stock	**P**	
81.	Quantity	**P**	
82.	Shipping Details	**P**	
83.	Connected Events	**P**	
84.	Transection	**P**	
85.	One-time Invitation	**P**	
86.	Submitted Request	**P**	
87.	Incoming event request	**P**	
88.	Current password	**P**	
89.	New Password	**P**	
90.	Forget Password	**P**	
91.	Pending List	**P**	
92.	Reason	**P**	
93.	Bargaining	**P**	
94.	Valid Ticket	**P**	
95.	Valid Payment	**P**	
96.	Role Based Feature	**P**	
97.	History	**P**	
98.	Full Access	**P**	
99.	Item	**P**	
100.	“pending” status	**S**	**1,13,14,15,20**

101.	System updates	**P**	
102.	Helpline	**P**	
103.	Time-sensitive OTP	**S**	**1,7**
104.	Booking System	**P**	
105.	Singer Profile(System View)	**P**	
106.	Instant Notification	**P**	
107.	Cart System	**P**	
108.	Checkout Process	**P**	
109.	“Used” Ticket Status	**P**	
110.	Reason	**P**	
111.	Extra Cost	**P**	
112.	Seller	**P**	
113.	Mandatory Fields	**P**	
114.	Total Amount	**P**	
115.	Intended Admin	**P**	
116.	Existing Admin	**P**	
117.	Strength Requirements	**P**	
118.	Pending List	**P**	
119.	Valid Complain	**P**	
120.	Duplicate Tickets	**P**	
121.	Available Items	**P**	
122.	Available Tickets	**P**	
123.	Platform Specifications	**P**	
124.	Login Page	**P**	
125.	Registration Page	**P**	
126.	Home Page	**P**	
127.	System Sends	**P**	
128.	System Displays	**P**	
129.	Account Suspension	**P**	
130.	Instant Notification	**P**	
131.	Live event status	**P**	
132.	QR scanner tool	**P**	
133.	Complain routing system	**P**	
134.	Automated Notification	**S**	**1,61**
135.	Tag event feature	**P**	
136.	OTP generation service	**P**	
137.	OTP verification Service	**P**	
138.	Internal Admin creation tool	**P**	
139.	Admin review list	**P**	
140.	Suspension reason field	**P**	
141.	Payment Prompt	**P**	

142.	Custom sub-domain buying option	**P**	
143.	Poster upload weight	**P**	
144.	Poster Upload date	**P**	
145.	Event Request Date	**P**	
146.	Event Request Time	**P**	
147.	Event Request Sender	**P**	
148.	Event Request Receiver	**P**	
149.	Total Request Send	**P**	
150.	Total Complain Submit	**P**	
151.	Complain Evidence	**P**	
152.	Add to cart	**P**	
153.	Delete from cart	**P**	
154.	Cart limit enforcer	**P**	
155.	Cart review page	**P**	
156.	Bulk payment processor	**P**	
157.	Complain form	**P**	
158.	Rejection	**P**	
159.	Sub-Domain Rejection	**P**	
160.	Modules	**P**	
161.	Interface	**P**	
162.	Queue	**P**	
163.	Validator	**P**	
164.	Width	**P**	
165.	Time	**P**	
166.	API	**P**	
167.	Algorithm	**P**	
168.	Engine	**P**	
169.	Full name	**P**	
170.	Browsing History	**P**	
171.	Summary	**P**	
172.	Transection	**P**	
173.	Time limit	**P**	
174.	Status	**P**	
175.	Oversight	**P**	
176.	Quantity	**P**	
177.	Pending List	**P**	
178.	Official ID	**P**	
179.	Fraud	**P**	
180.	Slot	**P**	
181.	Days	**P**	

182.	Month	**P**	
183.	Year	**P**	
184.	Special Character	**P**	
185.	Upper Letter	**P**	
186.	Lower Letter	**P**	
187.	Mixed	**P**	
188.	Submit	**P**	
189.	Show	**P**	
190.	View	**P**	
191.	T-shirts	**P**	
192.	Cap	**P**	
193.	Runtime	**P**	
194.	Error	**P**	
195.	Faild	**P**	
196.	Attack	**P**	
197.	Safe	**P**	
198.	Hamper	**P**	
199.	Demand	**P**	
200.	Low Demand	**P**	
201.	High Demand	**P**	
202.	High Price	**P**	
203.	Low Price	**P**	
204.	Peak time	**P**	
205.	Loss	**P**	

ER Diagram



Schema Table:

USER		
Attributes	Datatype	Size
U_id	Number	8
username	VAR_CHAR	20
email	VAR_CHAR	30
Profile picture	VAR_CHAR	20
Password	VAR_CHAR	20
Status	VAR_CHAR	20
Role	VAR_CHAR	20

ITEM		
Attributes	Datatype	Size
Seller_id	VAR_CHAR	20
Event_ID	VAR_CHAR	20
Type	VAR_CHAR	20
Name	VAR_CHAR	20
Description	VAR_CHAR	20
ITEM_ID	VAR_CHAR	20
Price	Number	8
Stock_Quantity	Number	8
QR_code	VAR_CHAR	20
used	VAR_CHAR	
Photo	VAR_CHAR	

QR Scan Log		
Attributes	Datatype	Size
Item_id	VAR_CHAR	20
Scan_id	VAR_CHAR	20
Scanned at	VAR_CHAR	30
Device	VAR_CHAR	20

ORDER		
Attributes	Datatype	Size
Shipping Name	VAR_CHAR	20
Total Amount	Number	8
Buyer id	VAR_CHAR	20
Order Id	VAR_CHAR	20
Shipping Address	VAR_CHAR	20
Shipping Phone	VAR_CHAR	20
Status	VAR_CHAR	20
Ordered At	VAR_CHAR	20

ORDER_ITEM

Attributes	Datatype	Size
Price	Number	8
Order_Item_id	VAR_CHAR	20
Order_id	VAR_CHAR	30
Item_id	VAR_CHAR	20
Quantity	Number	8

COMPLAINT		
Attributes	Datatype	Size
Suspect_u_id	VAR_CHAR	20
Event_id	VAR_CHAR	20
U_id	VAR_CHAR	20
Complaint_id	VAR_CHAR	20
Description	VAR_CHAR	20
Evidence	VAR_CHAR	20
Created_At	VAR_CHAR	20

Event		
Attributes	Datatype	Size
Fee	Number	8
Title	VAR_CHAR	20
Organizer_ID	VAR_CHAR	20
Tier_2 Quantity	Number	8
Status	VAR_CHAR	20
Singer_ID	VAR_CHAR	20
Event_ID	VAR_CHAR	20
Dynamic Ticket Pricing	VAR_CHAR	20
Description	VAR_CHAR	20
Tier_1 Price	Number	8
Poster	VAR_CHAR	20
Tier_2 Price	Number	8
Date	VAR_CHAR	20
Venue	VAR_CHAR	20
TIme	VAR_CHAR	20

CIty	VAR_CHAR	20
Tier_3 Price	Number	8
Tier_3 Quantity	Number	8
Tier_1 Quantity	Number	8
Description	VAR_CHAR	20

8. Class Cards:

1.User

Attributes	**Methods**
Username Email Role Password U_is Status Mobile Banking Number Performance Fee	+Registration() +Login() +Logout() +Update_Profile() +Got_Suspention() +Event_Request() +Submit_Complain() +Make_Payment() +Add_Product() +Place_Order()
Responsiblity	**Collaborator**
Event Perfomance Fee Payment	Payment
Event Request	User
Submit Complain	User
Buy Products	Payment
Launch	Event
Place	Order
Submit	Complain
Receives	Notification
Try to Verify	Verification

2.Event

Attributes	**Methods**

Id Title Singer_ID Organizer_ID Date Venue City Poster Description Total Ticket Quantity Sold Ticket Dynamic Ticket Pricing Created_At	+Create() +Launch() +Enable_Ticket_Pricing() +Buy_SubDomain() +Make_Payment()
Responsiblity	**Collaborator**
Event Request Send & Response	User
Event has some complain	Complain
Request Notification & Response Notification	Notification

3.Item

Attributes	**Methods**
Id Seller_ID Total_Quantity Available Quantity Description Tier Price Scanned Time	+Add_Stock() +Place_Order() +Enable_Dynamic_Pricing() +Buy_SubDomain() +Make_Payment()
Responsiblity	**Collaborator**
Valid QR code Scanning ,Sells	User
Belongs	Order

4.Payment

Attributes	**Methods**
Id User_ID Date Created_At	+Make_Payment()
Responsiblity	**Collaborator**
Paid By	User

5.Order

Attributes	**Methods**
Id User_ID Items Total_Amount	+Add_Item() +Remove_Item() +Update_Quantity() +Make_Payment()
Responsiblity	**Collaborator**
Place Order	Payment
Created	User

6.Complain

Attributes	**Methods**
User_Id Complain_ID Evidence Submitted_At Status	+File_Complaint() +Review_Complain()
Responsiblity	**Collaborator**
Against	User
Submit Complain	User

7.Verification

Attributes	**Methods**
Id User_ID Date OTP Status	+GenerateOTP() +VerifyOTP()
Responsiblity	**Collaborator**
Email Verification	User

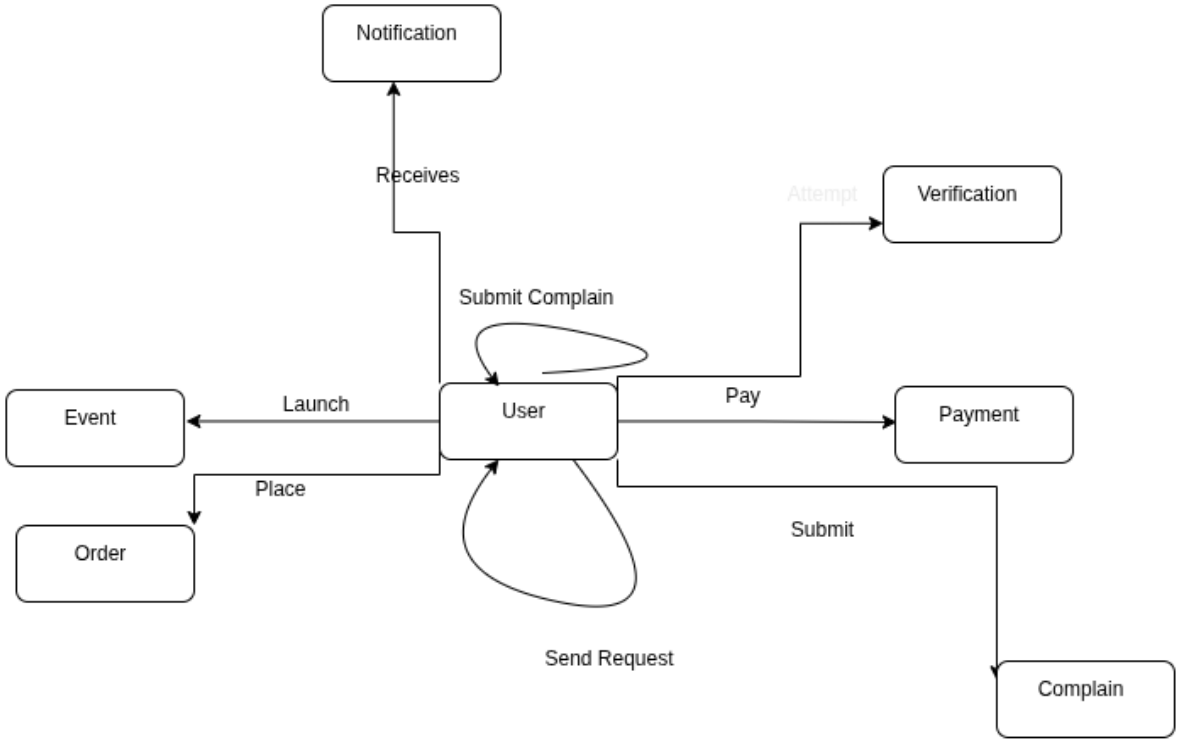
8.Notification

Attributes	**Methods**

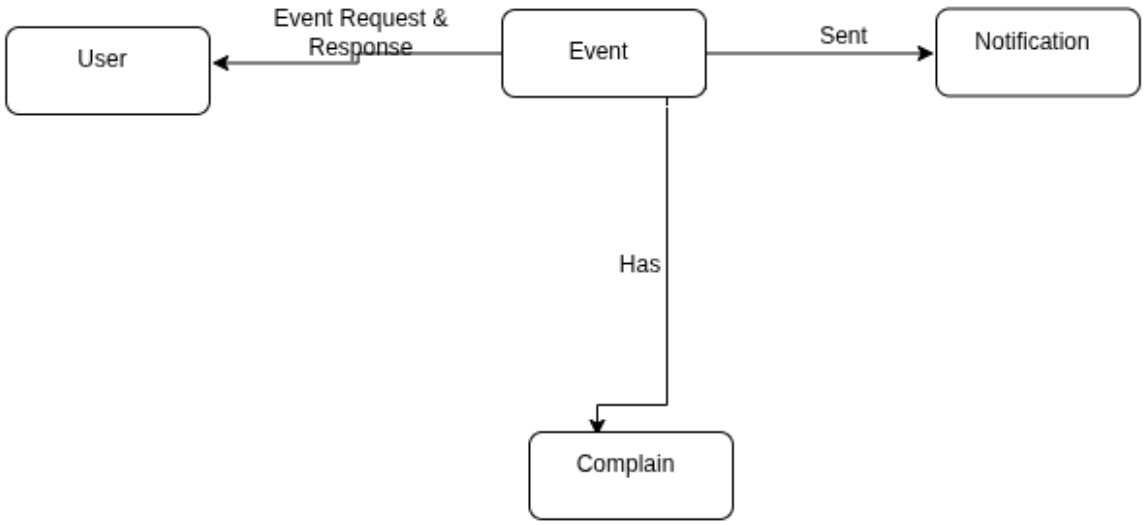
Id User_ID Message Status Created_At	+Send()
Responsiblity	**Collaborator**
Event Request Sending	User
Suspention Information	User

CRC Diagram

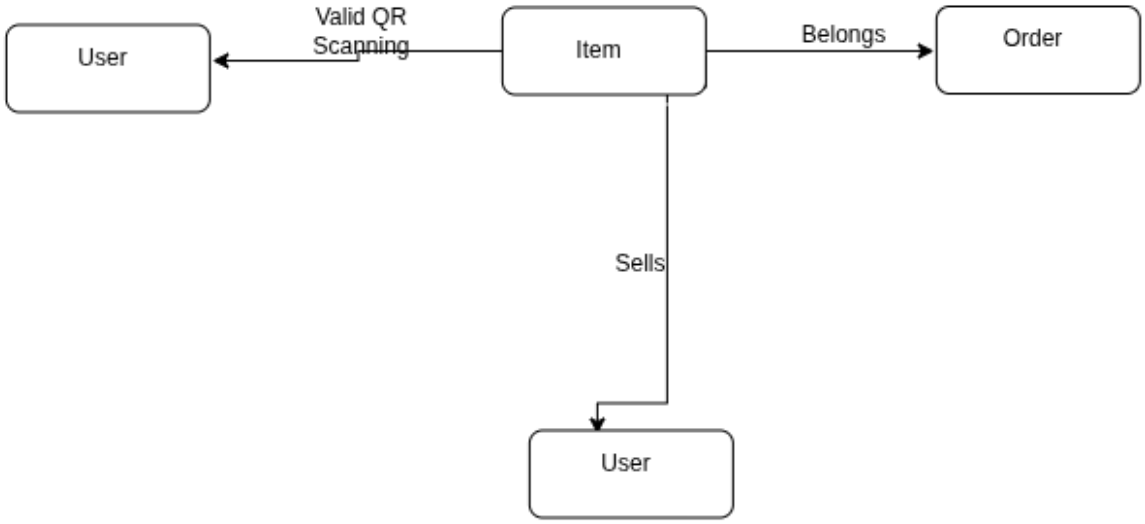
User



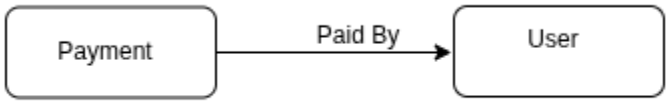
Event



Item



Payment

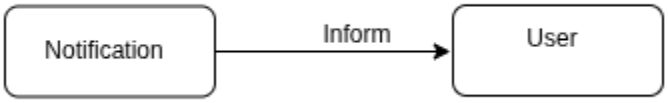


Order

Complain

Verification

Notification



9. Behavioral Modeling

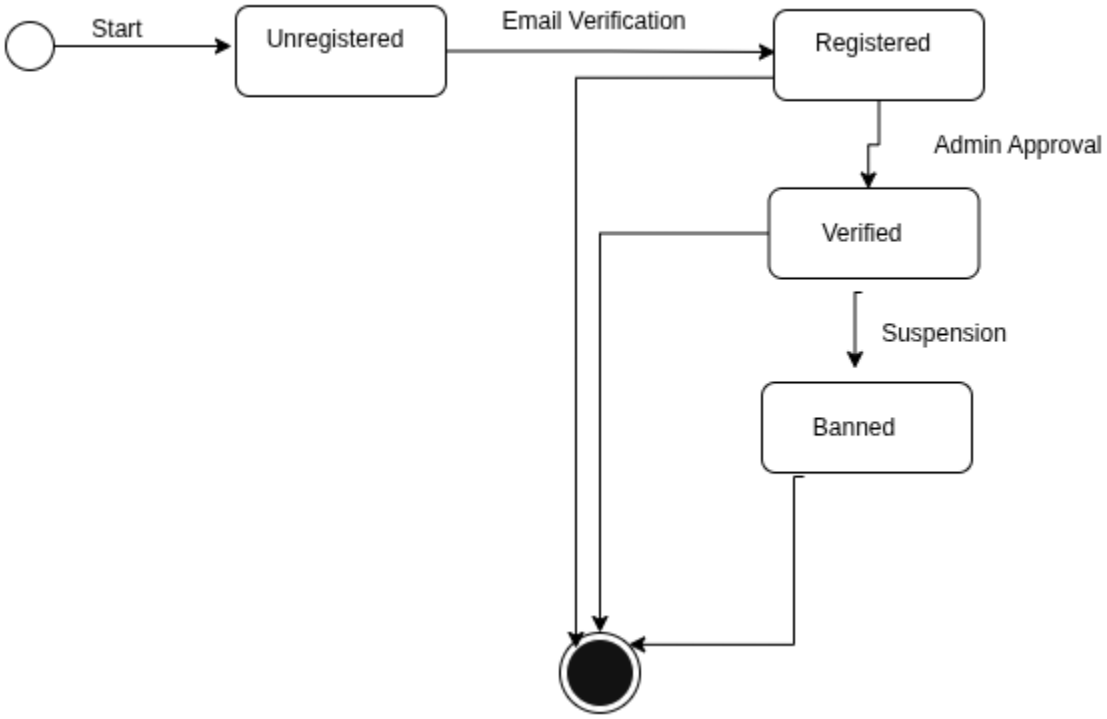
Serial	Event Name	Initiator	Collaborators
1	Registration	User	Verification
2	EmailAlreadyExist	System	User
3	OTPGenerated	System	User
4	OTPVerifiedSuccessfull	User	System
5	OTP Verified Failed	User	System
6	Account Created	User	System
7	Account Status Changed	System	User
8	Pending Account Notification	System	Admin
9	Admin Contacted User	Admin	User,Email
10	Identity Verified	Admin	User
11	Account Approved	Admin	User,System

12	Account Rejected	Admin	User,System
13	Admin Invitation	Admin (Existing)	System
14	Login Attempt	User	System
15	Login Successful	System	User
16	Login Failed	System	User
17	Pending Account Login Attempt	User	System
18	Password Reset Requested	User	System
19	Password Changed	User	System
20	Profile Updated	User	System
21	Mobile Banking Updated	User	System
22	Performance Fee Updated	Singer	System
23	Account Suspended	Admin	System,User
24	SingerBrowse	Organizer	System
25	SlotBookingRequested	Organizer	Singer
26	EventRequestReceived	System	Singer
27	EventRequestAccepted	System	Singer
28	EventRequestRejected	Singer	Organizer,System
29	EventFeePaid	Organizer	Singer,System
30	ConcertCreated	Organizer	System,Singer
31	CustomDomainRequested	Organizer	System,Admin
32	CustomDomainApproved	Admin	Organizer,System
33	ConcertLaunched	System	User
34	ConcertBrowse	Audience	System
35	TicketAddedToCart	User	System
36	CartCheckout	User	System
37	TicketPaymentSuccessful	Payment Gateway	Audience,QR code generator
38	TicketsDelivered	Email	User
39	QRCodeScanned	Audience	QR code Generator
40	TicketValidated	Audience	QR code Generator
41	TicketRejected	System	Organizer
42	ComplaintSubmitted	Audience	Organizer,Admin
43	StockAdded	Organizer/Singer	System
44	StockUpdated	Organizer/Singer	System
45	MarketplaceBrowse	User	System
46	ProductAddedToCart	User	System
47	CheckoutInitiated	User	System
48	ShippingDetailsConfirmed	User	System
49	OrderPlaced	User	System
50	OrderPaymentSuccessful	Payment Gateway	User

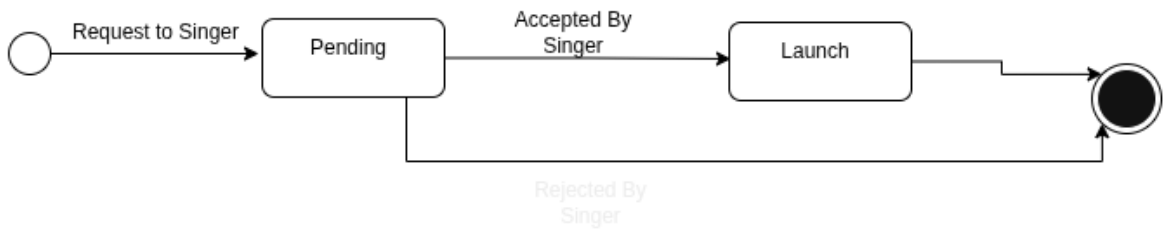
51	PendingAccountListViewed	Admin	Singers,Organizers
52	UserIdentityVerified	Admin	Singer,Organizer
53	DocumentsCollected	Admin	
54	ComplaintReviewed	Organizer	
55	ValidComplaintRegistered	Audience	Admin,Organizer
56	UserSuspendedDueToComplaints	Admin	Organizer,Singer
57	NotificationSent	Organizer	Singer
58	EmailSent	Admin	Singer,Organizer
59	PaymentProcessed	Organizer	Payment Gateway,Organizer
60	ConcurrentLoginDetected	Organizer	System

State Transition Diagram:

User:

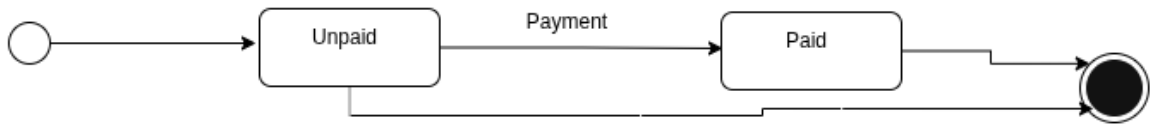


Event:

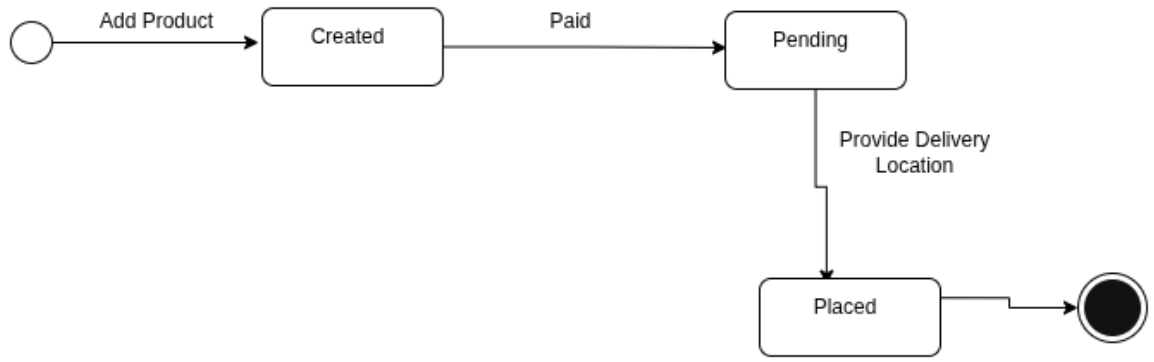


Item:

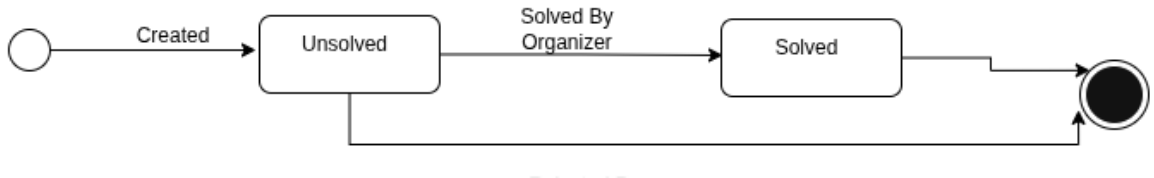
Payment:



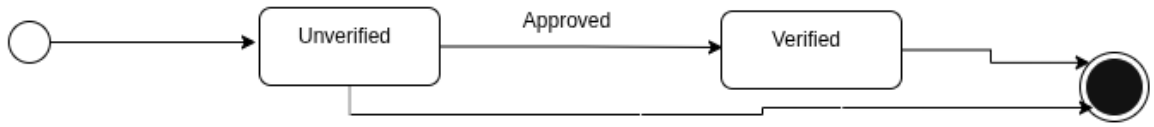
Order:



Complain:

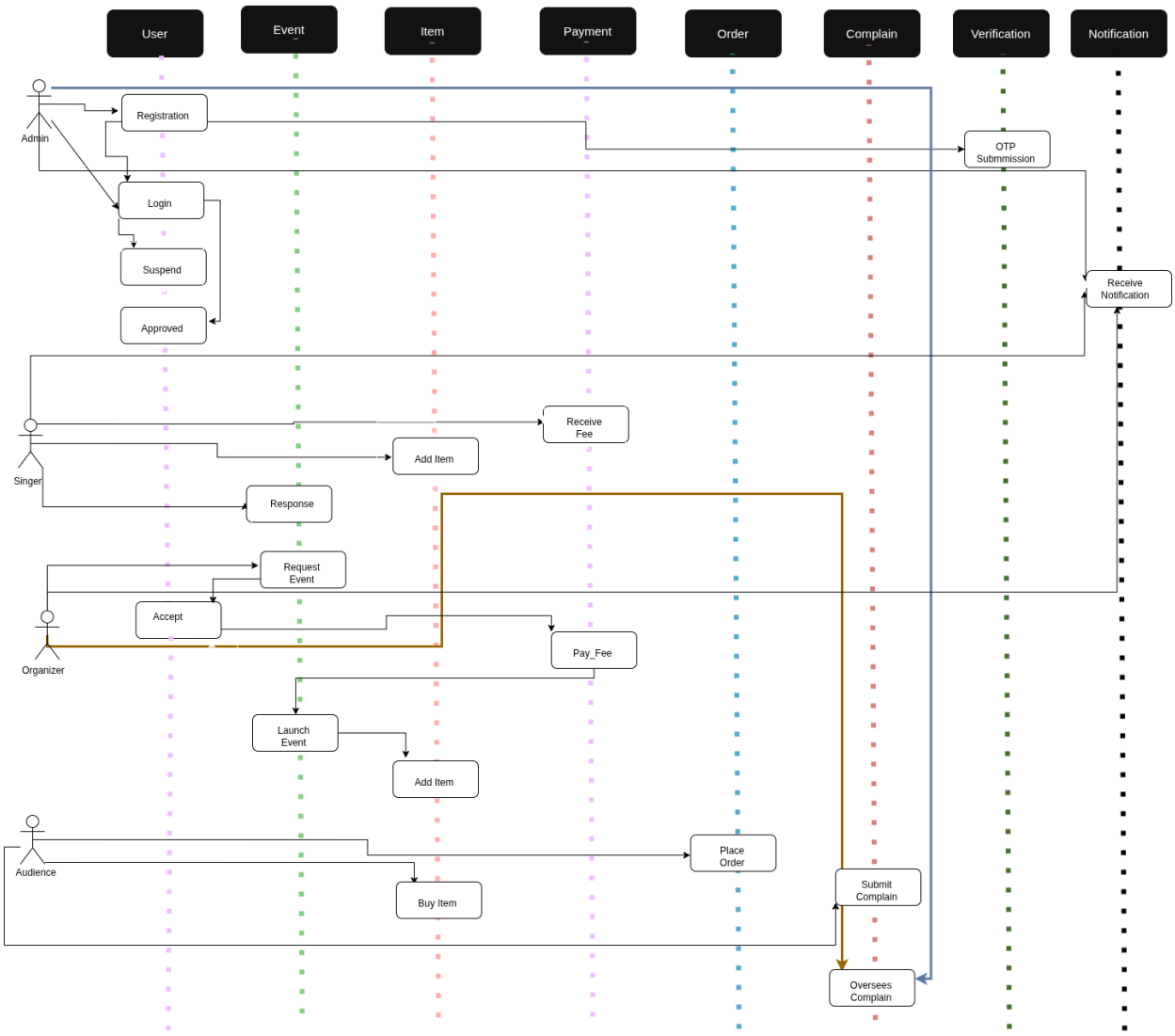


Verification:



Notification

10.Sequence Diagram


User
Event
Item
Payment
Order
Complain
Verification
Notification
Admin
Registration
OTP Submmission
Login
Suspend
Approved
Receive Notification
Singer
Receive Fee
Add Item
Response
Request Event
Accept
Organizer
Pay_Fee
Launch Event
Add Item
Audience
Place Order
Buy Item
Submit Complain
Oversees Complain